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## Shock tube study of ethanol pyrolysis II: Rate constant measurements and modeling

Rishav Choudhary<sup>a</sup>, Vivek Boddapati<sup>a</sup>, Sean Clees<sup>a</sup>, Julian J. Girard<sup>a</sup>, Yuzhe Peng<sup>b</sup>, Jiankun Shao<sup>a,\*</sup>, David F. Davidson<sup>a</sup>, Ronald K. Hanson<sup>a</sup><sup>a</sup> Department of Mechanical Engineering, Stanford University, Stanford CA-94305, United States<sup>b</sup> School of Aerospace Engineering, Georgia Institute of Technology, Atlanta GA-30332, United States

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## ABSTRACT

The rate constants of two unimolecular decomposition channels of ethanol,  $\text{C}_2\text{H}_5\text{OH}(\text{+Ar}) = \text{CH}_3 + \text{CH}_2\text{OH}(\text{+Ar})$ , and  $\text{C}_2\text{H}_5\text{OH}(\text{+Ar}) = \text{C}_2\text{H}_4 + \text{H}_2\text{O}(\text{+Ar})$  were measured near 1 atm and 10 atm behind reflected shock waves between 1190 and 1550 K by tracking the evolution of CO and  $\text{C}_2\text{H}_4$ , respectively. Use of sensitive laser diagnostics in conjunction with carefully selected test mixtures enabled the suppression of sensitivity of CO and  $\text{C}_2\text{H}_4$  time-histories to secondary reactions, resulting in a significant reduction of uncertainty in the measured rate constants. Additionally,  $\text{CH}_3\text{CHO}$  time-histories measured during the pyrolysis of 2%  $\text{C}_2\text{H}_5\text{OH}/\text{Argon}$  mixtures were used to determine the rate constant of the most dominant H-abstraction channel,  $\text{C}_2\text{H}_5\text{OH} + \text{H} = \text{sC}_2\text{H}_4\text{OH} + \text{H}_2$  between 1290 and 1530 K. The low temperature  $\text{CH}_3\text{CHO}$  time-histories were also used to infer the rate constant of  $\text{sC}_2\text{H}_4\text{OH} = \text{CH}_3\text{CHO} + \text{H}$ . The measured rate constants were used to generate an updated kinetic model, whose performance was evaluated against the species time-history data reported in recent shock tube pyrolysis studies. The performance of the updated model was also evaluated against ignition delay times (IDTs) of stoichiometric  $\text{C}_2\text{H}_5\text{OH}/\text{Oxygen}/\text{Nitrogen}$  mixtures, measured between 1000 and 1500 K near 20 atm in this study. The updated kinetic model was found to show excellent agreement with both shock tube pyrolysis and IDT studies. Additionally, the reduction in the uncertainty of the rate constants studied in this work resulted in a significant reduction in the uncertainty of the model predictions. The rate constant expressions listed below were derived from the measurements in this work, and are valid in the temperature range of 1200–1600 K.

$$k_{\text{C}_2\text{H}_5\text{OH}(\text{+Ar})} \leftrightarrow \text{CH}_3 + \text{CH}_2\text{OH}(\text{+Ar}), 1 \text{ atm} = 2.125 \times 10^{13} e^{-\frac{34070}{T}} \text{ s}^{-1} (\pm 21\%)$$

$$k_{\text{C}_2\text{H}_5\text{OH}(\text{+Ar})} \leftrightarrow \text{CH}_3 + \text{CH}_2\text{OH}(\text{+Ar}), 10 \text{ atm} = 3.079 \times 10^{14} T^{0.23} e^{-\frac{38030}{T}} \text{ s}^{-1} (\pm 28\%)$$

$$k_{\text{C}_2\text{H}_5\text{OH}(\text{+Ar})} \leftrightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}(\text{+Ar}), 10 \text{ atm} = 6.062 \times 10^{36} T^{-6.45} e^{-\frac{42180}{T}} \text{ s}^{-1} (\pm 31\%)$$

$$k_{\text{sC}_2\text{H}_4\text{OH}} \leftrightarrow \text{CH}_3\text{CHO} + \text{H}, 1 \text{ atm} = 5.817 \times 10^9 e^{-\frac{11372}{T}} \text{ s}^{-1} (\pm 55\%)$$

$$k_{\text{C}_2\text{H}_5\text{OH} + \text{H}} \leftrightarrow \text{sC}_2\text{H}_4\text{OH} + \text{H}_2 = 1.780 \times 10^7 T^{1.94} e^{-\frac{1835}{T}} \text{ s}^{-1} (\pm 28\%)$$

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## 1. Introduction

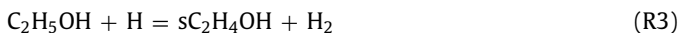
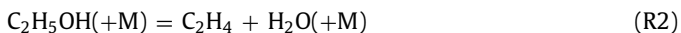
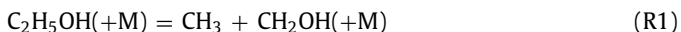
The exhaustive study of ethanol pyrolysis reported in [1] highlighted critical deficiencies in the existing chemical kinetic models for ethanol. None of the five kinetic models considered in that work could simultaneously reconcile the measured time-histories of all species. Of the five models considered, Aramco 3.0 [2] was found to show the closest agreement with the mea-

sured time-histories of all key products and intermediates. However, systematic differences between the measurements and model predictions were observed. Specifically, Aramco was found to underpredict both CO and  $\text{C}_2\text{H}_4$  time-histories at 1 atm and 10 atm and was found to overpredict  $\text{CH}_3\text{CHO}$  time-histories at 1 atm between 1190 and 1550 K. The mismatch between the measurements and model predictions hint towards inaccuracies in the rate constants of key reactions. Addressing these deficiencies is critical as the chemical kinetics of ethanol is foundational to the models describing kinetics of real transportation fuels.

\* Corresponding author.

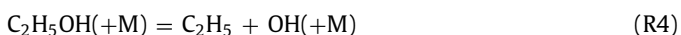
E-mail address: [jkshao@stanford.edu](mailto:jkshao@stanford.edu) (J. Shao).

Sensitivity and rate-of-production (ROP) analysis highlighted the importance of reactions R1, R2, and R3 in dictating the evolution of species time-histories. The extent of discrepancies between the model and the measurements was also found to vary with pressure. These observations emphasize the need for: 1) re-evaluation of the rate constants  $k_1$ ,  $k_2$ , and  $k_3$  of reactions R1, R2, and R3, 2) study of pressure dependence of the rate constants of R1 and R2, and 3) reduction in the overall uncertainty of the rate constants.



Theoretical and experimental studies directed towards determining the rate constants of R1-R3 have been relatively scarce. Herzler et al. [3] reported the first measurements of the rate constant of the overall ethanol decomposition reaction ( $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{Products}$ ). The rate constant was determined using ethanol and ethylene yields measured in a shock tube using a GC, between 1200 and 1300 K and 1–2 bar and were primarily intended to supplement the modeling of tetraethoxysilane pyrolysis. It was assumed that R1 and R2 were the only possible channels of decomposition and that the reactions followed second-order kinetics. Theoretical and experimental studies that followed have established that both of these assumptions are untrue [4–7]. However, there is a consensus that R1 and R2 are the most dominant decomposition channels [4–7]. Marinov [4] reported the first theoretical calculation of  $k_1$  and  $k_2$  based on the solution of the RRKM/Master kinetic equation using the hindered Gorin model. The calculated  $k_2$  showed good agreement with the measurements of Herzler et al. at 2 atm. However, the calculated  $k_1$  was found to be 70% slower than the measurements.

Park et al. [5] calculated *ab initio* rate constants for eight possible channels of thermal decomposition of ethanol and identified that R1 remains the dominant channel of decomposition at pressures below 10 atm. At higher pressures, and at temperatures above 1500 K, R2 becomes dominant and a third channel, R4, also becomes competitive.



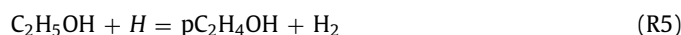
The calculated rate constant of R1 near the high-pressure limit,  $k_{1,\infty}$ , was in good agreement with the measurements reported by Park et al. [8], which were taken in a shock tube using a CO laser absorption diagnostic, and in a static cell using FTIR spectroscopy. However, the Park et al. rate was an order of magnitude slower than the Marinov rate. In his calculations, Marinov treated the three decomposition channels (R1, R2, and R4) as independent reactions. Tsang [9] argued that in multi-channel systems, the magnitude of energy transfer per collision and the differences in the rates of depletion of the excited reactant molecule across different channels would affect the eventual branching ratio of each channel. This effect is more pronounced at low pressures where the collision frequency is smaller. At these conditions, Marinov's assumption of treating each of the channels as independent reactions does not hold. Tsang's calculated rate constants consequently show a shallower falloff behavior for both  $k_1$  and  $k_2$ .

Li et al. [6] measured  $k_2$  in a flow reactor between 1045 and 1080 K in the pressure range of 1.7–3 atm. Their measurements show good agreement with the measurements of Herzler et al. and the theoretical calculations of Tsang et al. Li et al. also performed theoretical calculations of  $k_1$  and  $k_2$  by taking tunneling effects into account. Sivaramakrishnan et al. [7] performed an exhaustive

experimental and theoretical study of rate constants relevant to ethanol pyrolysis. They used OH absorption spectroscopy and H-ARAS to measure the rate constants of the three decomposition channels, and H-abstraction by OH and H. However, their study was limited to pressures lower than 1.5 atm. While their theoretical calculations of  $k_1$  are more than 3 times faster than those of Li et al. and Park et al. at 1250 K, and 1 atm, their calculated  $k_2$  shows good agreement with the calculations of Li and Park. Wu et al. [10] also measured  $k_1$  in a shock tube using H-ARAS in the pressure and temperature range of 1–2 atm and 1450–1760 K and found a reasonably good agreement with the measurements of Sivaramakrishnan et al. More recently, Kiecherer et al. [11] and Pinzon et al. [12] have reported measurements of  $k_2$  near 1 atm. Their measurements are consistent with the theoretical calculations of Li et al. [6], and Park et al. [5], and with the measurements of Li et al.

The difficulty in identification of optimal experimental conditions for measuring these rate constants, and the lack of powerful diagnostic tools have prevented further additions and improvements to the existing knowledge base for  $k_1$  and  $k_2$ . The experimental investigations have been limited to a narrow pressure range of 0.2–3 atm, which lies in the falloff regime of both  $k_1$  and  $k_2$ . Moreover, the uncertainty factor of the rate constants reported by most of these studies is far greater than 1.5. This affords a very small dataset to modelers to validate their theoretical calculations against. Consequently, the reported theoretical rate constants for both reactions at the high- and low-pressure limits show significant disagreements spanning several orders of magnitude. The difficulty in modeling is further compounded by the fact that Sivaramakrishnan et al., Li et al., and Wu et al. used different bath gasses in their studies. Sivaramakrishnan et al. used krypton as the diluent in their experiments, while Li et al. and Wu et al. used nitrogen, and argon, respectively. Given the scatter and uncertainty associated with these studies, it is difficult to ascertain the differences in chaperon efficiencies of these diluents. As noted by Shao et al. [13] and Choudhary et al. [14], the identity of the diluent molecules can have a substantial impact on the falloff behavior of the rate constants.

There have been significantly fewer studies of the H-abstraction channel,  $\text{C}_2\text{H}_5\text{OH} + \text{H} = \text{Products}$ , in the reported literature. The first experimental measurement of this rate constant was reported by Bansal and Freeman [15], who used  $\gamma$ -radiolysis of  $\text{C}_2\text{H}_5\text{OH}-\text{C}_3\text{H}_6$  mixtures at 423 K to determine the overall rate constant. Aders and Wagner [16] measured the rate constants in the temperature range of 295–700 K in a stirred reactor using mass spectrometric measurements of reaction products. These studies assumed that all H-abstraction sites are similar and consequently did not report branching ratios for various possible channels. Park et al. [17] performed theoretical calculations of the rate constants of four possible channels of H-abstraction. In addition to R3, they considered reactions R5–R7 in their calculations.



The overall rate constant calculated by Park et al. [16] was 70% slower than the measurements of Aders and Wagner [16] but was in good agreement with the Bansal and Freeman [15] data. The calculations of Park et al. [17] revealed that R3 is the dominant H-abstraction channel in the temperature range of 500–2500 K. In the temperature range of 1200–1500 K, the branching ratios of R3, R5, and R6 are 0.75, 0.2, and 0.05, respectively. Reaction R7 contributes negligibly. Sivaramakrishnan et al. [7] measured the overall rate constant of  $\text{C}_2\text{H}_5\text{OH} + \text{D} = \text{Products}$  in the temperature range

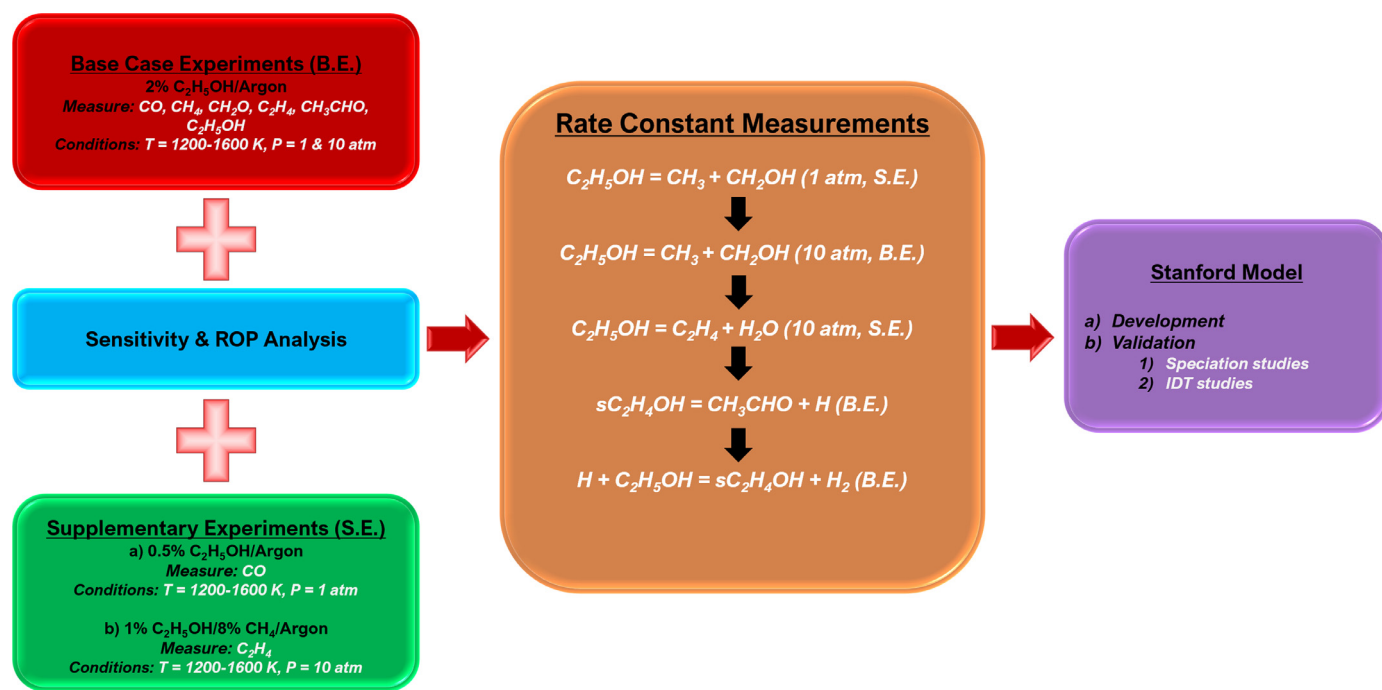


Fig. 1. Schematic of the work presented in [1] and this paper. Base case experiments were reported in [1]. Rate constant studies, supplemental experiments, and model development and validation are discussed in this paper. The rate constants were determined sequentially in the order shown in the figure.

of 1054–1359 K. They used DD-ARAS with shock heated C<sub>2</sub>H<sub>5</sub>OH/C<sub>2</sub>D<sub>5</sub>I/Kr mixtures to distinguish between the H-radicals generated by consumption of ethanol and the precursor DD-radicals formed upon decomposition of C<sub>2</sub>D<sub>5</sub>I. They performed theoretical calculations to quantify the isotope effect on the overall reaction rate (reported to be less than 5%). Theoretical calculations were also performed to quantify the branching ratios of individual channels. The measured rate constants were slower than theoretical calculations of Sivaramakrishnan et al. but were faster than the extrapolation of the calculations of Park et al. To date, the work of Sivaramakrishnan et al. [7] remains the only study to report high-temperature measurements of the rate constants of C<sub>2</sub>H<sub>5</sub>OH + D/H → Products.

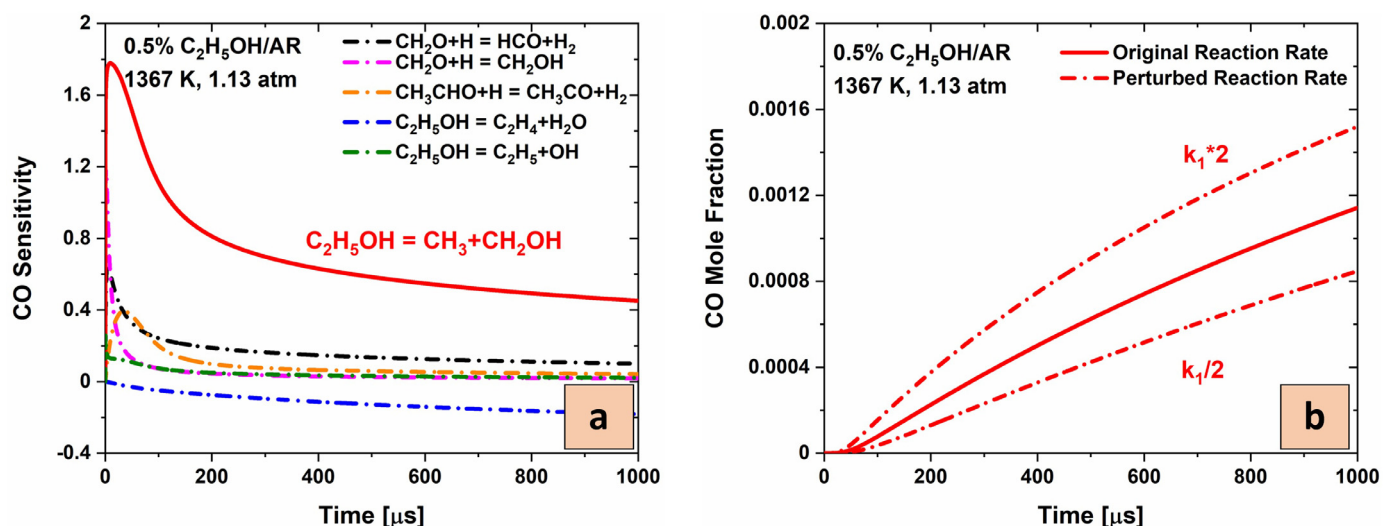
Of the five kinetic models considered in [1], Aramco 3.0 and Ranzi et al. [18] use the rate constants suggested by Sivaramakrishnan et al. [7] for ethanol decomposition and H-abstraction by H and OH, while the Saxena and Williams [19] model uses the rate constants suggested by Li et al. [6]. The model of Olm et al. [20] was developed by an optimization procedure that tuned the rate constants used in the Saxena and Williams model to match a range of IDT, flame speed, and species yield data measured in flow reactors. The Marinov model uses the rate constants that were theoretically calculated by Marinov [4]. Thus, the rate constants measured and calculated by Sivaramakrishnan et al., Li et al., and Marinov are essentially the only rate constants that have been adopted by kinetic modelers. As discussed above, these rate constants differ quite significantly from each other, have not been validated beyond a small range of conditions, and have undesirably high uncertainty factors. The goals of this work are motivated by these deficiencies in the current state of knowledge. The objectives are threefold: 1) Measure rate constants of reactions R1-R3 with tight uncertainty bounds, with a focus on addressing the pressure dependence of  $k_1$  and  $k_2$ , 2) Assess the performance of a modified kinetic model containing the updated rate constants against the shock tube species time-history data reported in [1], and against other speciation, and IDT data, 3) Assess the reduction in the uncertainty of the species time-histories predicted by the updated model. The flowchart shown in Fig. 1 gives an outline of the approach taken in this study

to address these three goals. Time-histories reported in [1] are supplemented with additional experiments, which are then used for determining the rate constants of key reactions. The measured rate constants are then used for developing an updated kinetic model (referred to as the Stanford model) for ethanol pyrolysis and oxidation. Finally, the performance of the updated model is compared against multiple pyrolysis and IDT studies reported in literature.

## 2. Experimental setup & methods

### 2.1. Shock tubes & laser diagnostics

The Kinetics Shock Tube (KST) and the High-Pressure Shock Tube (HPST) facilities at Stanford were used in this study. In addition to the experiments reported in [1], a total of 8 experiments were conducted on the KST while a total of 9 experiments were conducted on the HPST. The details of both facilities have been described in [1] and are omitted here for brevity. High-purity (>99.5%), anhydrous ethanol procured from Sigma-Aldrich was used in the experiments. High-purity helium, methane and argon for the rate constant experiments, and high purity synthetic air (21% O<sub>2</sub>/79% N<sub>2</sub>) used in the IDT experiments, were provided by Praxair. The mixtures were manometrically prepared, magnetically stirred and stored in heated stainless-steel mixing tanks for 30 min prior to each experiment. Pre-shock absorbance of the test mixture recorded at 3.41 μm was used in conjunction with the cross-section of pure ethanol at 20°C and 90°C at 3.41 μm (measured *a priori*) to infer the fuel loading in each shock experiment on KST and HPST, respectively. The average fuel loss across all shock experiments was 2%. Laser absorption spectroscopy was used to measure the evolution of the mole fractions of key species during each experiment. The procedure for inferring mole fraction from the attenuation of the incident laser intensity during the experiment has been already described in [1]. The additional experiments described in this work were used for inferring  $k_1$  at 1 atm and  $k_2$  at 10 atm. In conjunction with a subset of measurements reported in [1],  $k_1$  at 10 atm, and  $k_3$  were also determined. The measured CO time-histories were used for determining  $k_1$ , while C<sub>2</sub>H<sub>4</sub> time-



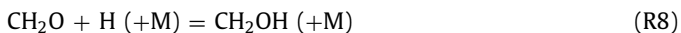
**Fig. 2.** (a) Sensitivity plot showing the top six reactions to which the time-history of CO forming during the pyrolysis of 0.5% C<sub>2</sub>H<sub>5</sub>OH/Ar at 1367 K, 1.13 atm is sensitive. The simulations were performed using ARAMCO 3.0. (b) CO time-histories simulated using Aramco 3.0 with the unmodified mechanism, and with the rate constant of R1 multiplied and divided by 2.

histories were used for inferring  $k_2$ . The details of their respective diagnostics are described in [1].

## 2.2. Rate constant determination methodology

Sensitivity analysis was used to identify the optimal test conditions for inferring the rate constants of interest. Aramco 3.0 was used as the base mechanism for performing sensitivity calculations. The selection of test conditions was guided by three considerations: 1) The species to be measured should exhibit high and exclusive sensitivity to the target rate constant, 2) The signal-to-noise ratio (SNR), and consequently the absorbance of the target species, at the test conditions should be high, and 3) The temperature change during the experiment, stemming from the endothermicity of the reaction should be low. These considerations are imperative to reduce the uncertainty in the rate constants determined using the time-histories.

Fig. 2(a) shows the sensitivity plot of CO time-histories during the pyrolysis of a 0.5% C<sub>2</sub>H<sub>5</sub>OH/Argon mixture at 1367 K, 1.13 atm. The plot suggests that the CO time-histories are almost exclusively sensitive to R1 at these conditions. The contribution of secondary reactions, dominated by R8 and R9, diminishes after the first 30  $\mu$ s.



Between 30–800  $\mu$ s, the only significant contribution to the sensitivity is from R9, which has been recently measured by Wang et al. [21, 22], who quote an uncertainty factor of 1.1. The high sensitivity of CO to R1 is further confirmed by Fig. 2(b) where the nominal prediction of the CO time-history by Aramco is plotted against the predictions obtained by perturbing the rate constant of R1 by a factor of 2. Therefore, the logical procedure to infer  $k_1$  is to perturb the rate constant until the residual of the simulated and the measured CO time-history at this condition is minimized. This procedure was used to infer the  $k_1$  in the temperature range of 1234–1504 K. The fitting interval for each experiment was guided by the sensitivity analysis. The lower bound of the interval was set at the instant when the sensitivity to the most dominant interfering reaction reduced to less than one-fourth of the sensitivity to R1, while the upper bound of the fitting interval was chosen as the instant when the absolute sensitivity of CO time-histories to R1 dropped to 0.4. The interval ranged from 30 to 800  $\mu$ s at 1234 K

to 0–300  $\mu$ s at 1504 K. The measured absorbance time-histories in all experiments were converted to mole fractions using Beer's law and simulated temperature time-histories. In simulating the temperature evolution, the pyrolysis process was assumed to occur at a constant pressure; an assumption supported by the measured pressure traces. Owing to the minimal temperature change, and the relative insensitivity of the absorption cross-sections to temperature, the inferred mole fraction traces are expected to have a negligible dependence on the thermodynamic constraints used to model these experiments.

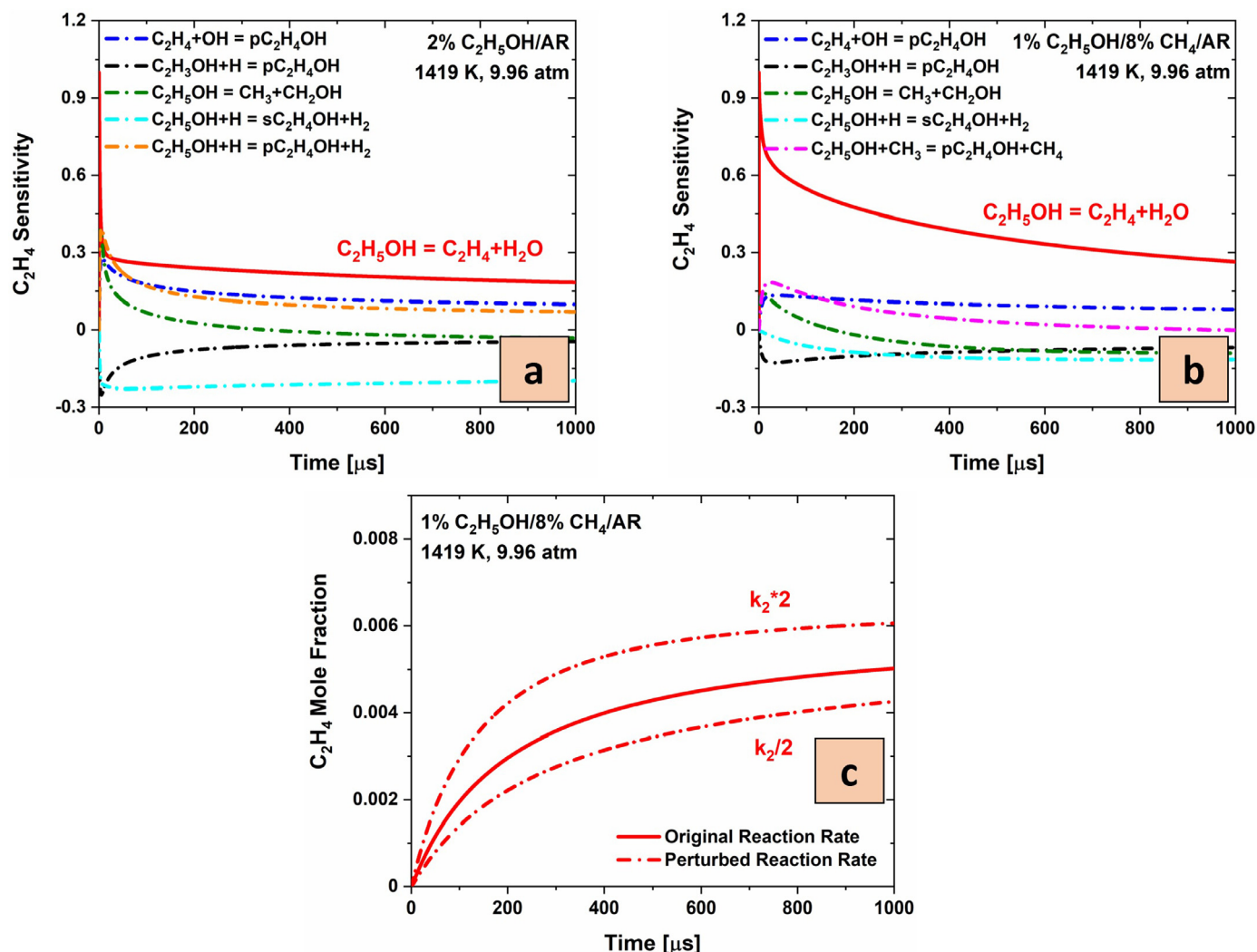
The sensitivity of ethylene time-history during the pyrolysis of a 2% C<sub>2</sub>H<sub>5</sub>OH/Argon mixture at 10 atm is shown in Fig. 3(a). While ethylene does exhibit high sensitivity to R2, there is significant interference from other reactions, dominated by R3. A radical scavenger is needed to suppress the sensitivity of ethylene to R3 and other bimolecular reactions involving H radicals. Methane was used as a radical scavenger in this study. The chemical kinetics of methane is well-understood, and methane is thermally stable in the range of temperatures studied in this work. Hence, it serves as an excellent radical scavenger. The optimal test conditions that provide a high sensitivity to R2 while minimizing the influence of secondary reactions between 1200 and 1500 K at 10 atm, were found to require 1% C<sub>2</sub>H<sub>5</sub>OH/8% CH<sub>4</sub>/Argon as the test mixture. The suppression of sensitivity of ethylene time-histories to secondary reactions for the aforementioned mixtures is illustrated in Fig. 3(b). Fig. 3(c) shows the nominal prediction of the ethylene time-history by Aramco 3.0 at one of the experimental conditions and the predicted time-histories obtained by perturbing the rate constant of R2 by a factor of 2. Akin to the procedure described above for obtaining  $k_1$ , the best fit  $k_2$  was obtained by adjusting the rate constant until the residual of the simulated and measured ethylene time-histories was minimized. The procedure to determine the rate constants described in this section is general and was repeated with the measured CH<sub>3</sub>CHO time histories to infer  $k_3$ , and the rate constant of thermal decomposition of sC<sub>2</sub>H<sub>4</sub>OH.

## 3. Results and discussions

### 3.1. Rate constant measurements

The results of the rate constant studies, and detailed uncertainty analysis for each of the reactions is described in their respective sub-sections that follow.





**Fig. 3.** (a) Sensitivity plot showing the top six reactions to which the time-history of  $C_2H_4$  forming during the pyrolysis of 2%  $C_2H_5OH/Ar$  at 1419 K, 9.96 atm is sensitive. Notice that R2 is not the most dominant reaction. (b) Sensitivity plot showing the impact of addition of a radical scavenger,  $CH_4$  on  $C_2H_4$ . Notice the simultaneous enhancement in sensitivity to R2 and the reduction in the sensitivity to other reactions. (c)  $C_2H_4$  time-histories simulated using Aramco 3.0 with the unmodified mechanism, and with the rate constant of R2 multiplied and divided by 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

### 3.1.1. $C_2H_5OH = CH_3 + CH_2OH$

Fig. 4 shows two example measurements of CO time-histories at 1367 K and 1440 K. Also shown are the simulated best-fit time-histories, and the simulated time-histories obtained by perturbing the best fit rate constants by 20%. The difference in the measured and simulated CO time-history at each time-step, normalized by the measured CO concentration at that time-step was used to calculate the residual. Minimization of the sum of the squares of the residuals over the fitting interval was chosen as the metric to determine the best-fit rate constant. The uncertainty in the rate constants determined using this procedure can stem from multiple sources, i.e. uncertainty in absorption cross-section of CO, uncertainty in the fitting procedure, uncertainty in time-zero of the reaction, uncertainty in the initial reaction temperature, mixture uncertainty, and the influence of secondary reactions. The overall 2- $\sigma$  uncertainty was calculated by computing the root-mean-squared (RMS) sum of the uncertainty stemming from each of the individual sources listed above.

The 2- $\sigma$  uncertainty in the absorption cross-section of CO, obtained from [23] was propagated using Beer's law to obtain the uncertainty bounds on the measured CO time-histories. The fitting procedure described in Section 2.2 was repeated with the upper-

and lower-time-history bounds substituting the measured CO time-histories. The best-fit rate constants thus obtained represent the contribution of the uncertainty in the absorption cross-section to the overall rate constant uncertainty. This procedure was repeated for quantifying the uncertainty contributions from the choice of time-zero by shifting the measured time-histories forwards and backwards in time by 5  $\mu s$ . The choice of the magnitude of time-shift reflects the average rise time of the pressure transducer used in the experiments, which was nearly 10  $\mu s$ . The post-shock 2- $\sigma$  temperature uncertainty (0.7% for the current experiments), was propagated using the Arrhenius equation (obtained by fitting the rate constants inferred from individual experiments to their respective initial temperature), to quantify its contribution to the rate constant uncertainty. The impact of mixture uncertainty was quantified by repeating the fitting procedure with modified simulated traces obtained by perturbing the initial mixture composition by 2%, which represents the average discrepancy between the manometrically prepared and spectroscopically measured initial fuel loading.

The Monte Carlo sampling method described by Shao et al. [24, 25] was followed to quantify the contribution of secondary reactions to the overall uncertainty. Table S1 in the Supplementary

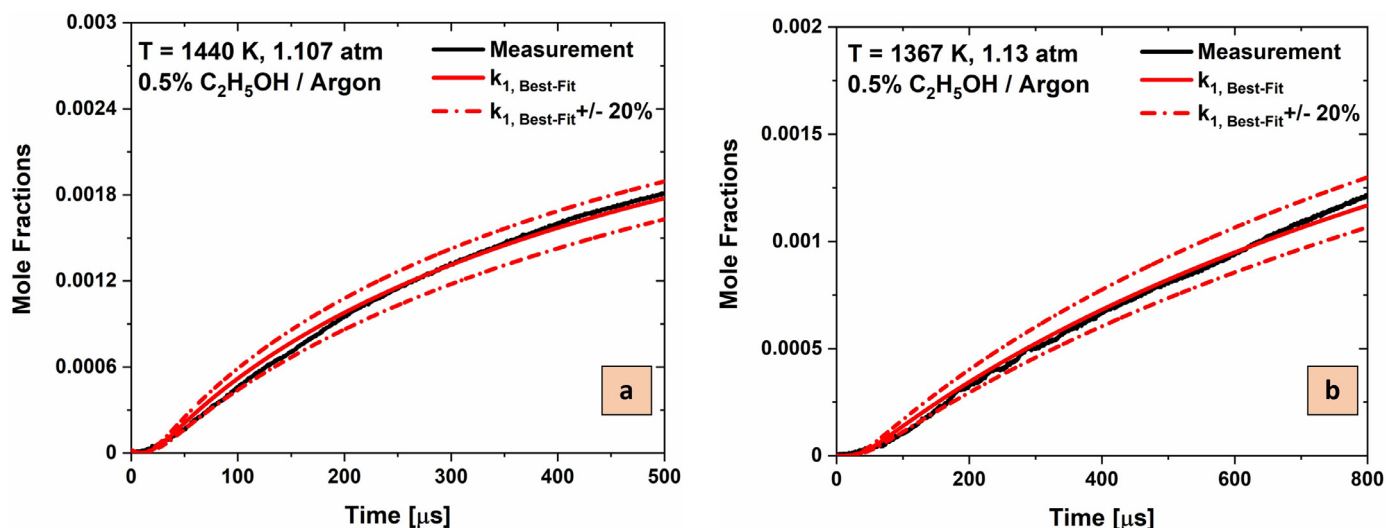


Fig. 4. (a), (b) Measured and best-fit CO time-histories for two example cases. The dash-dotted lines represent the CO time-histories simulated by perturbing the best-fit rate constant by 20%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

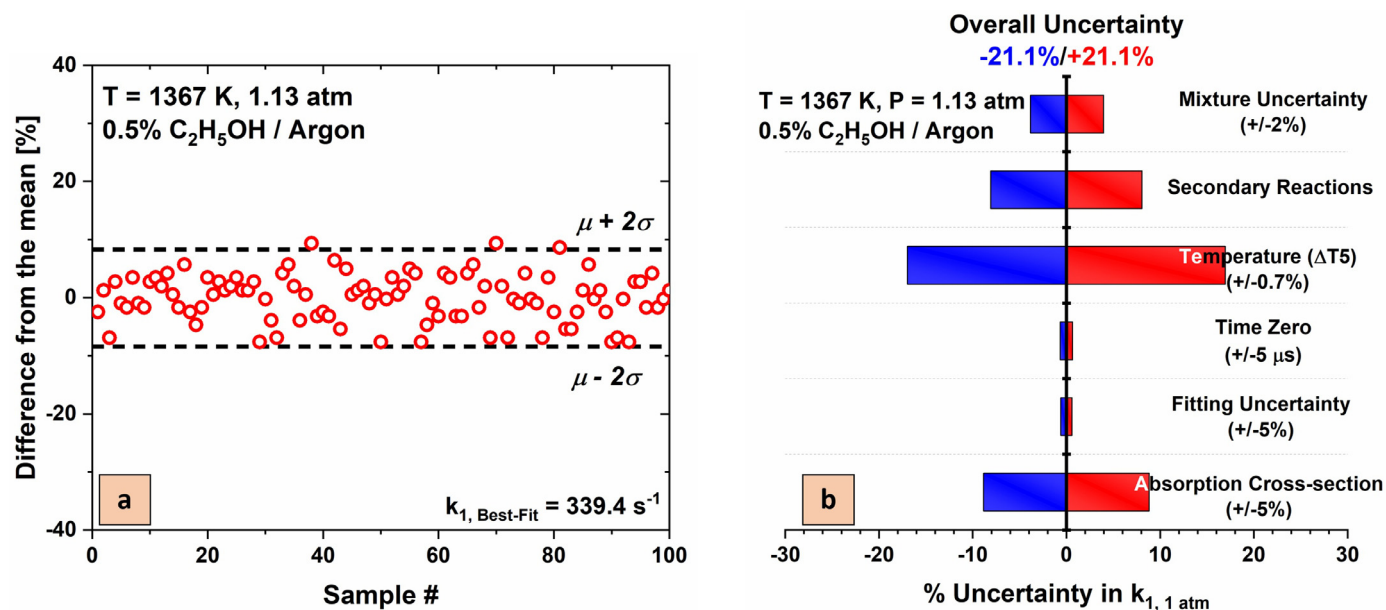


Fig. 5. (a) Results of the Monte Carlo sampling procedure used for inferring the contribution of secondary reactions to the overall uncertainty of  $k_1$  for the test case shown in fig. 4(b). Each symbol represents the percentage deviation of the best-fit rate constant obtained using a modified ARAMCO 3.0 model from the best-fit rate constant obtained using the unmodified ARAMCO 3.0 model. (b) Uncertainty analysis for the measured  $k_1$  for the test case shown in fig. 4(b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Material lists seven reactions, besides R1, to which the CO time-histories are most sensitive. Also listed in table S1 are the corresponding uncertainty factors and the references. The method entails determination of the best-fit rate constant,  $k_1$ , by repeating the aforementioned fitting procedure but starting with a different base kinetic model ("sampled" model). A sampled model is generated by multiplying the Arrhenius A-parameter of each of the seven influencing reactions by a randomly chosen factor. This factor is constrained to lie within the uncertainty factors of the respective reaction. Hundred such sampled models were generated and used with the fitting routine. The procedure resulted in a set of hundred best-fit rate constants unique to each model. These best-fit rate constants lie within a narrow interval that also encompasses the  $k_1$  obtained using Aramco 3.0 as the base model. The interval containing 95% these rate constants was identified and was used to calculate the 2- $\sigma$  uncertainty in  $k_1$  due to the influ-

ence of secondary reactions. The scatter plot (Fig. 5(a)) shows the best fit rate constants obtained for each sample model. The contribution of each of these sources to the overall uncertainty for one of the experiments is graphically shown in Fig. 5(b). The overall 2- $\sigma$  uncertainty is +21.1%/−21.1%. Uncertainty in post-shock temperature contributes the most to the overall uncertainty, followed by the uncertainty in the absorption cross-section and the influence of secondary reactions. Fitting uncertainty and time-zero uncertainty have a negligible contribution to the overall uncertainty.

The rate constants calculated from each experiment are shown as a function of temperature on an Arrhenius plot (Fig. 6). The measurements exhibit significantly lower scatter and uncertainty than previous investigations. The measured rate constants are faster than the Sivaramakrishnan et al. [6] rate constant at all temperatures. The difference between the two rates increases with temperature which suggests that the Sivaramakrishnan et al. rate

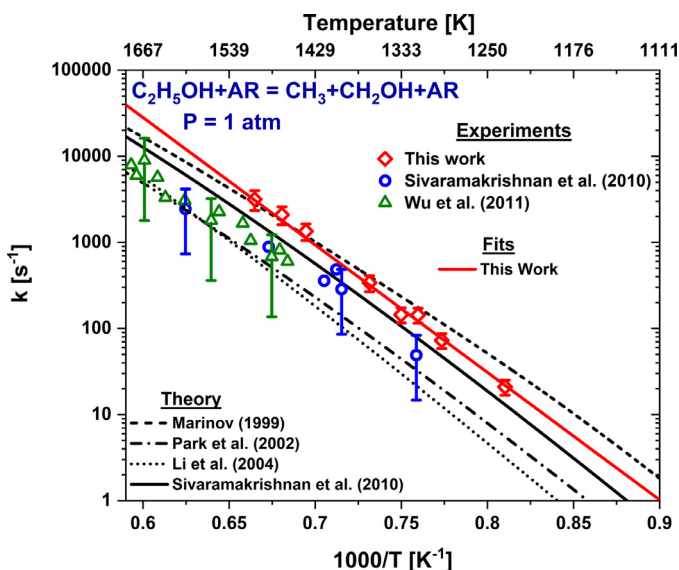


Fig. 6. Arrhenius plot for the rate constant of  $C_2H_5OH = CH_3 + CH_2OH$  at 1 atm.

constant has a smaller activation energy. It is worth noting that Sivaramakrishnan et al. did not directly measure the rate constant of  $k_1$  but used the initial slope of the OH time-histories measured during ethanol decomposition with a fitting procedure to jointly determine the branching ratios of different decomposition channels. Similarly, Wu et al. [10] used separate experiments to infer the branching ratio of this channel and the overall rate of thermal decomposition. The measured rate constants are also faster than those theoretically calculated and measured by Park et al. [4] and Li et al. [5] but are slower than the theoretical calculations of Marinov. Fitting the measured rate constants to temperature yields the Arrhenius expression given by equation 1.

$$k_{1,1 \text{ atm}} = 2.125 \times 10^{13} \exp\left(-\frac{34070}{T}\right) \text{ s}^{-1} \quad (1)$$

The procedure described in the preceding paragraphs was also followed for determining the rate constant of R1 at 10 atm using the CO time-histories measured during pyrolysis of 2%  $C_2H_5OH$ /Argon mixtures near 10 atm (reported in the companion

paper [1]). The time-histories measured during pyrolysis of ethanol at 10 atm are also dominantly sensitive to R1 (Fig. 7(a)). However, the absolute sensitivity of CO time-histories to secondary reactions is greater at 10 atm than at 1 atm. An example high-pressure measurement of CO time-history is shown in Fig. 7(b). The first 500  $\mu\text{s}$  of the measured time-histories were used with the fitting algorithm to determine the best-fit rate constant. Also shown are the best-fit simulated time-history along with the time-histories obtained by perturbing the best-fit rate constant by its corresponding overall uncertainty,  $\pm 30\%$ . The overall uncertainty was calculated using the procedure described earlier. Comparison of the scatter plots in Figs. 5(a), and 8(a) suggest that the secondary reactions result in a higher overall uncertainty at 10 atm than at 1 atm, which is commensurate with the greater sensitivity of CO time-histories to secondary reactions at 10 atm. The overall uncertainty in the rate constant is  $+28.4\%/-27.2\%$  and is again dominated by the uncertainty in post-shock temperature, influence of secondary reactions and the uncertainty in absorption cross-section (Fig. 8(b)).

The rate constants determined using time-histories measured in the temperature range of 1200 K – 1600 K are plotted against temperature on an Arrhenius plot shown in Fig. 9. These measurements exhibit significantly lower scatter, tighter uncertainty bounds, and show good agreement with the measurements of Park et al., which remains the only previous experimental study to date to report measurements of  $k_1$  at 10 atm. Akin to the low-pressure measurements, the measured rate constants are systematically larger than the theoretical calculation of Sivaramakrishnan et al. [7], Li et al. [6], and Park et al. [5], but smaller than the rate constant calculated by Marinov [4]. Fitting the measured rate constants to temperature yields a non-Arrhenius type expression given by equation 2.

$$k_{1,10 \text{ atm}} = 3.079 \times 10^{14} T^{0.23} \exp\left(-\frac{38030}{T}\right) \text{ s}^{-1} \quad (2)$$

The rate constants determined from each experiment are listed in Table A1 in the Appendix along with their respective uncertainties. The pressure dependence of  $k_1$  is illustrated in Fig. 10. The rate constants given by Eqs. (1) & (2) are plotted against temperature along with the current measurements, and those reported by other groups in literature on the same Arrhenius plot. It is observed that the ratio of rate constants measured at 10 atm and 1 atm increases with temperature (from  $\sim 3$  at 1250 K, to  $\sim 6$  at

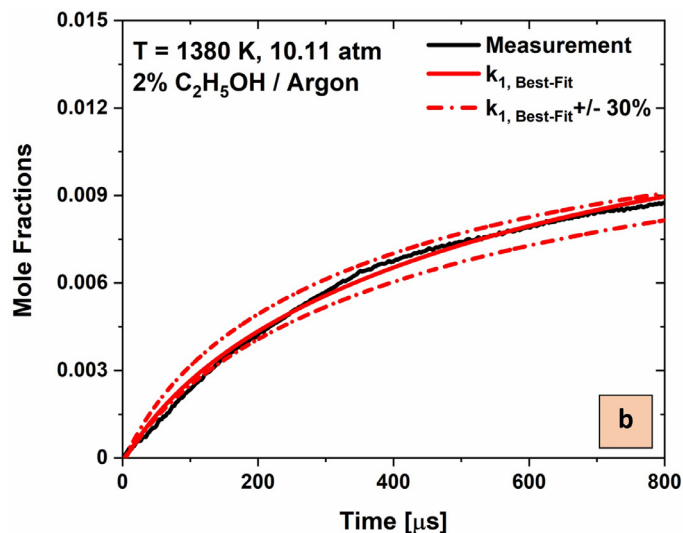
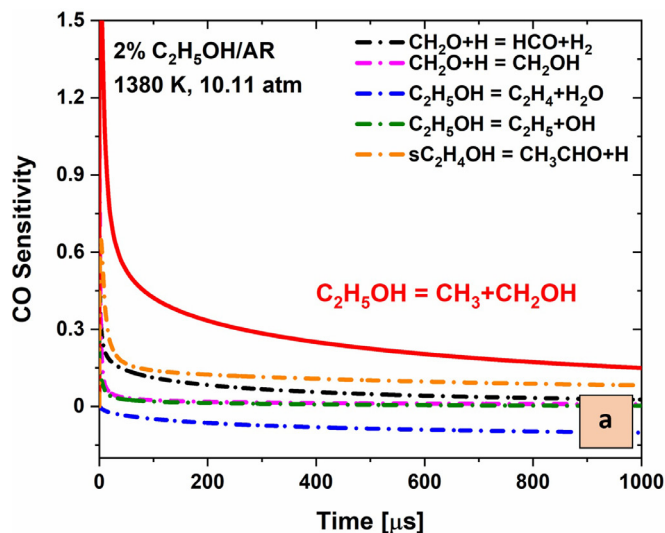
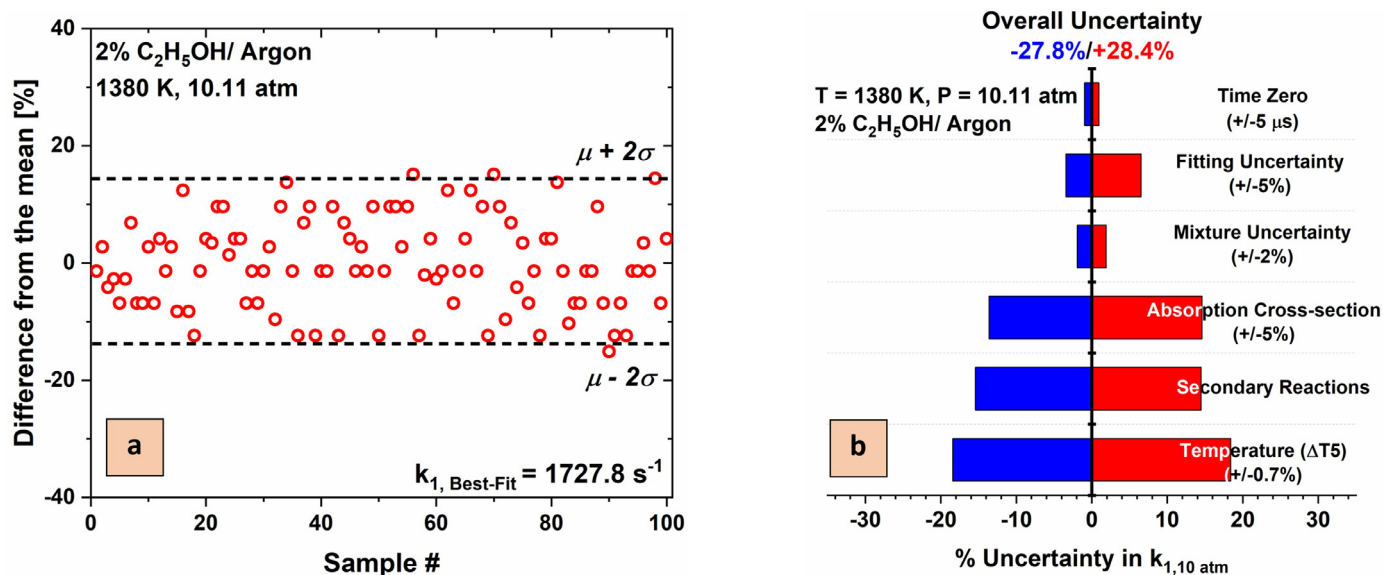


Fig. 7. (a) Sensitivity of CO time-history during the pyrolysis of a 2%  $C_2H_5OH$ /Argon mixture at 1380 K, 10.11 atm. (b) Measured and best-fit CO time-histories. The dash-dotted lines represent the CO time-histories simulated by perturbing the best-fit rate constant by 30%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)





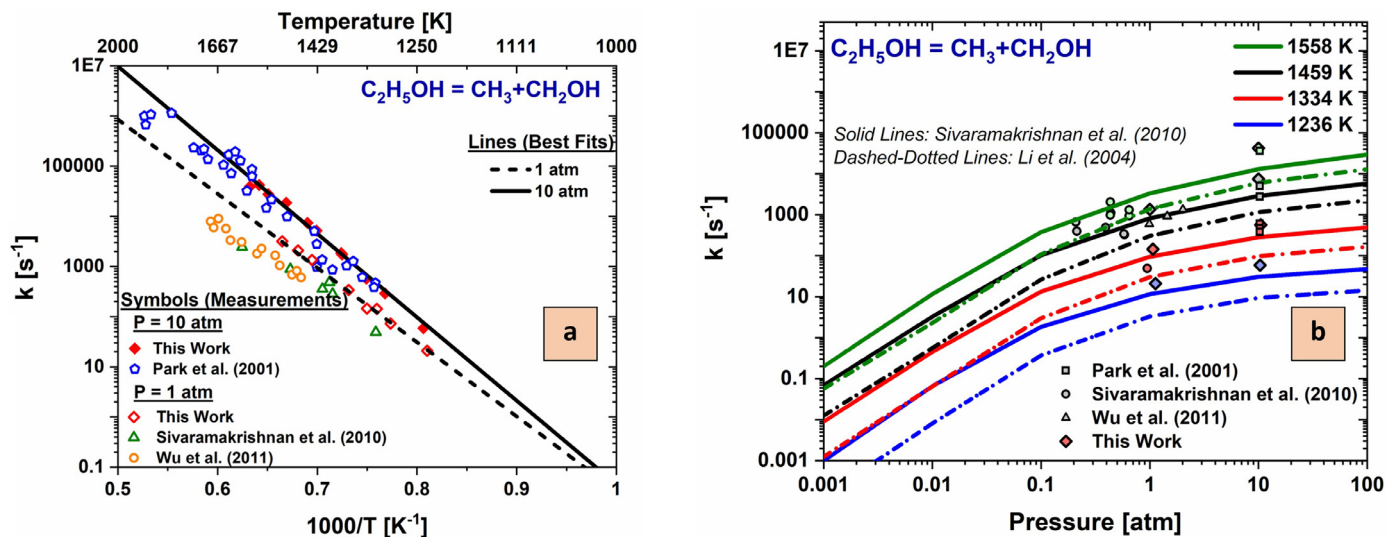
**Fig. 8.** (a) Results of the Monte Carlo sampling procedure used for inferring the contribution of secondary reactions to the overall uncertainty of  $k_{1,10 \text{ atm}}$  for the test case shown in Fig. 6(a). (b) Uncertainty analysis for the measured  $k_{1,10 \text{ atm}}$  for the test case shown in Fig. 6(a). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

1500 K). The pressure dependence is further emphasized by falloff curves, shown for four temperatures in the right panel of Fig. 10. The falloff curves were generated using the theoretically calculated rate expressions of Sivaramakrishnan et al. [7], and Li et al. [6]. These curves suggest that in the temperature range spanned by the experiments in this work, the rate constants approach their high-pressure limit at pressures significantly greater than 100 atm. The plot also suggests that all experimental studies of  $k_1$  have been conducted in the intermediate pressure (or falloff) regime. While the present study has improved our knowledge of  $k_1$  by enabling the direct determination of  $k_1$  at 1 atm and 10 atm within tight uncertainty bounds, further improvement warrants measurement of  $k_1$  over a wider range of pressures. Future studies should focus on the measurement of  $k_1$  at higher pressures. These studies will further improve our knowledge of the pressure dependence of  $k_1$  and will enable the development of improved theoretical models.

### 3.1.2. $\text{C}_2\text{H}_5\text{OH} = \text{C}_2\text{H}_4 + \text{H}_2\text{O}$

The rate constant of R2 near 10 atm was determined using ethylene time-histories measured during the pyrolysis of 1% C<sub>2</sub>H<sub>5</sub>OH/8% CH<sub>4</sub>/Argon mixtures between 1200 and 1600 K. As described in Section 2.2, use of methane sensitized test mixtures renders the measured ethylene time-histories dominantly sensitive to R2, while simultaneously suppressing the sensitivity to other reactions. The procedure followed to determine rate constants and overall uncertainty from the measured time-histories, have been described in the preceding sub-sections. A representative time-history comparing the measured and best-fit simulated time-histories is shown in Fig. 11.

The contribution of secondary reactions to the overall uncertainty was determined using the same Monte Carlo sampling method that was followed for  $k_1$ . The list of reactions considered in the Monte Carlo sampling along with their rate constants and uncertainty factors are listed in Table S2 of the Supplementary



**Fig. 10.** (a) Arrhenius plot of the best-fits to the  $k_1$  measured in this work at 1 and 10 atm compared against current and historical data. (b) Universal falloff curves for R1 calculated using the Sivaramakrishnan et al. [7] and Li et al. [6] expressions at four different temperatures. Symbols represent current and historical measurements of the rate constant. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



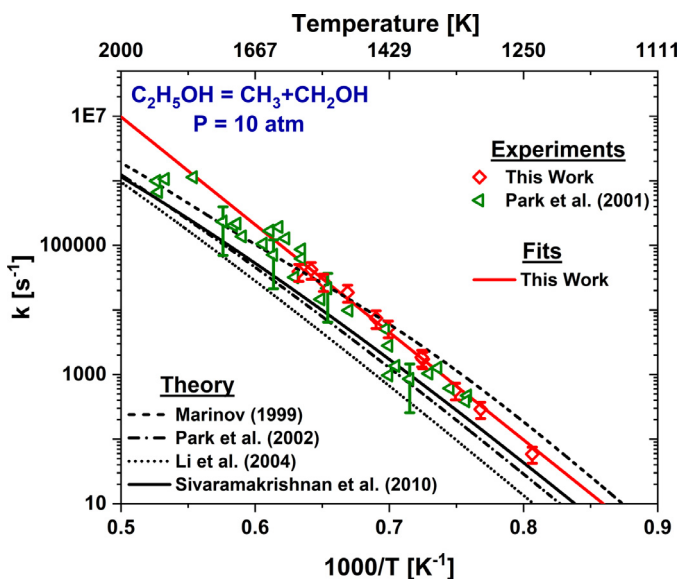
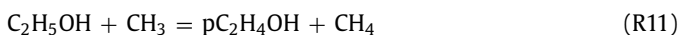


Fig. 9. Arrhenius plot for the rate constant of  $C_2H_5OH = CH_3 + CH_2OH$  at 10 atm.

Material. At the selected experimental conditions, the secondary reactions are dominated by R10 and R11.



The rate constant of R10 is relatively well known. The most recent study of R10 was undertaken by Lockhart et al. [26] with a quoted uncertainty factor of 1.2. Fewer studies of R11 are available in the literature. Most kinetic models for ethanol use the rate constants theoretically calculated by Park et al. [27]. While their calculations show good agreement with the rate constants measured at low temperatures, no experimental studies have followed to confirm the validity of their calculations at higher temperatures. Therefore, an uncertainty factor of 2 was assumed for R11 in the present work. Other reactions that were considered for uncertainty quantification include R1, R3, and the second decomposition chan-

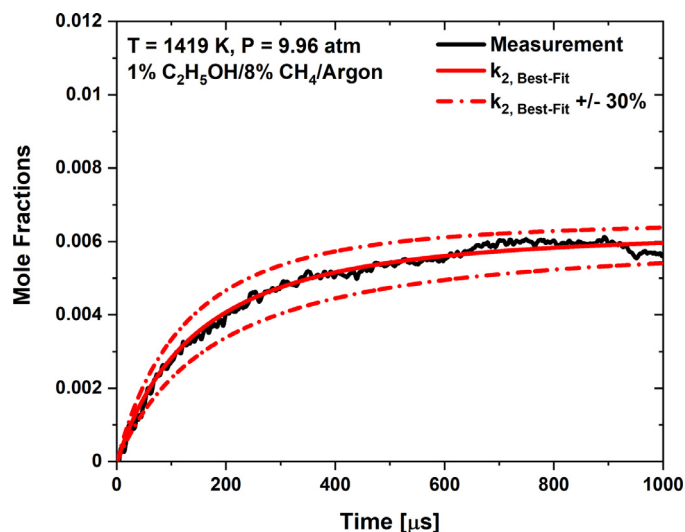


Fig. 11. Measured and best-fit  $C_2H_4$  time-histories for one representative condition. The dash-dotted lines represent the  $C_2H_4$  time-histories simulated by perturbing the best-fit rate constant by 30%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

nel of the  $pC_2H_4OH$  radical. The rate constant of R1 has been accurately determined in this work with an uncertainty factor of 1.2 (Section 3.1.1). Uncertainty factors of the remaining secondary reactions, along with their respective references are listed in table S2 of the Supplementary Material. Despite the high uncertainty factors associated with the secondary reactions, proper selection of experimental conditions in this work has ensured that their overall contribution to the uncertainty is small ( $\sim 20\%$  at the representative test case, as shown in Fig. 12(a)). The results of the uncertainty analysis for the example case are shown in Fig. 12(b). The overall uncertainty in  $k_2$  is  $+31.1\%/-30.9\%$  for the representative case, with the uncertainty in post-shock temperature, absorption cross-section, and secondary reactions contributing almost equally.

The measured rate constants are plotted against temperature and shown on an Arrhenius plot in Fig. 13. Also plotted are the

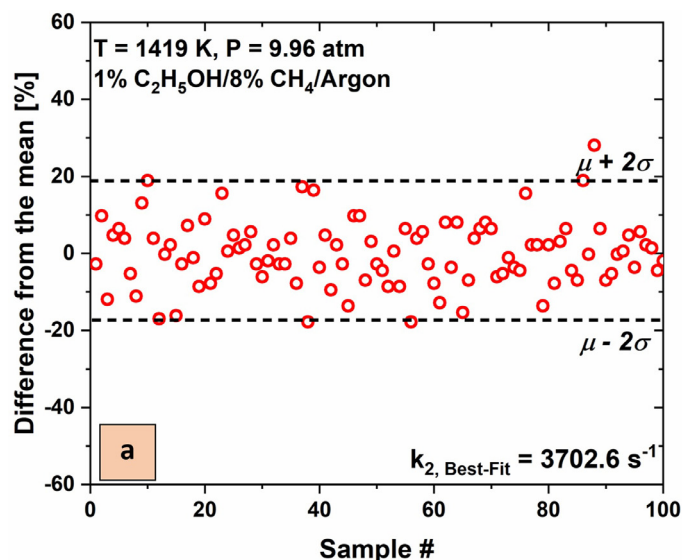
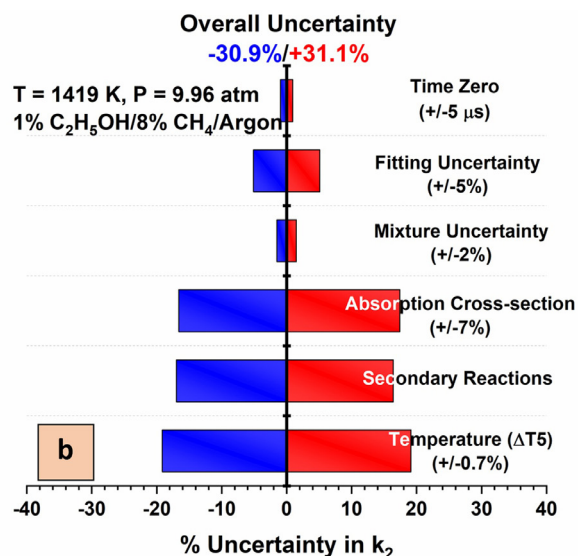


Fig. 12. (a) Results of the Monte Carlo sampling procedure used for inferring the contribution of secondary reactions to the overall uncertainty of  $k_{2,10 \text{ atm}}$  for the test case shown in Fig. 10. (b) Uncertainty analysis for the measured  $k_{2,10 \text{ atm}}$  for the test case shown in Fig. 10.



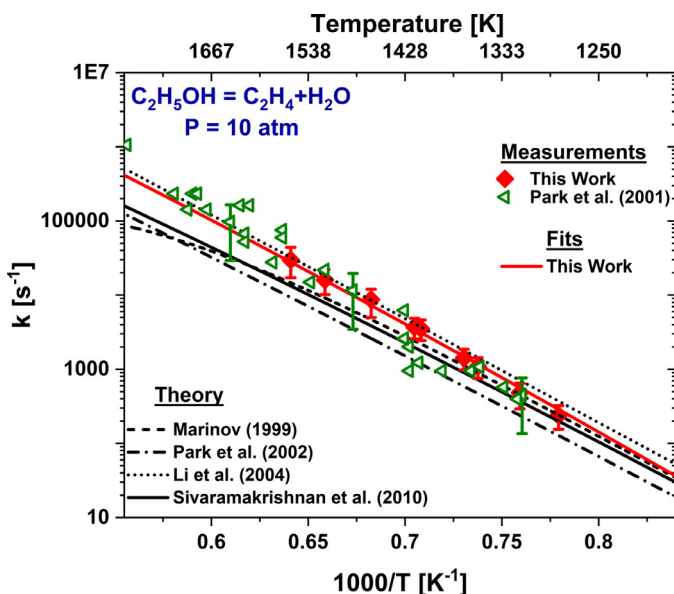


Fig. 13. Arrhenius plot for the rate constant of  $C_2H_5OH = C_2H_4 + H_2O$  at 10 atm.

rate constants computed using the theoretical rate expressions calculated by Marinov [4], Park et al. [5], Li et al. [6], and Sivaramakrishnan et al. [7], and the experimental measurements of Park et al. [8]. The current measurements show good agreement within the scatter of the measurements of Park et al. [8]. The measurements also exhibit significantly lower scatter than the Park et al. [8] data. The measured rate constants are faster than the theoretical calculation of Marinov, Park et al., and Sivaramakrishnan et al., but slower than the Li et al. rate over the entire temperature range spanned by the experiments conducted in this work. Fitting the measured rate constants to temperature yields a non-Arrhenius type expression given by equation 3.

$$k_{2,10 \text{ atm}} = 6.062 \times 10^{36} T^{-6.45} \exp\left(-\frac{42180}{T}\right) s^{-1} \quad (3)$$

The Arrhenius plot in Fig. 14(a) effectively summarizes the updated state of knowledge of  $k_2$ . While high-pressure measurements of  $k_2$  are scarce, numerous studies in the recent past have reported mea-

surements of  $k_2$  at 1 atm. Fitting the 1 atm rate constants to temperature gives equation 4.

$$k_{2,1 \text{ atm}} = 1.922 \times 10^{47} T^{-9.60} \exp\left(-\frac{45020}{T}\right) s^{-1} \quad (4)$$

The rate constant expressions given by Eqs. (3) & 4 reconcile with all experimental measurements in the temperature range of 1000–2000 K. Contrary to R1, R2 shows a weaker pressure dependence. For example, at 1250 K, the ratio of rate constants at 10 atm and 1 atm for R2 is 2. At the same temperature, the ratio of 10 atm to 1 atm rate constant for R1 is 3. The falloff curves generated using Sivaramakrishnan et al. [7], and Li et al. [6] rate constants for R2 are shown in Fig. 14(b) at four temperatures. The curves suggest that at 1280 K, the lowest temperature at which the experiments were conducted in this work for inferring  $k_2$ , the rate constant is very close to its high-pressure limit at 10 atm. Again, most of the experimental studies reported in literature aimed at studying R2 have been conducted in the falloff regime. Future studies aimed towards measuring the high-pressure limit of  $k_2$  will enable further improvement in the theoretical modeling of  $k_2$ .

The impact of the updated rate constants on the branching ratios of thermal decomposition is shown in Fig. 15. The rate constants of R1, given by Eqs. (1) & 2, and R2, given by Eqs. (3) & 4 were used in conjunction with the Sivaramakrishnan et al. [7] rate constant for the third decomposition channel, R4, for the calculation of the branching ratios. Fig. 15 confirms that R1 and R2 are the dominant channels of unimolecular decomposition. R2 is the primary decomposition pathway at temperatures below 1400 K at both 1 atm and 10 atm. The updated reaction rate constants suggest that the crossover in reactivity occurs near 1800 K and 1400 K at 1 atm and 10 atm, respectively. The crossover occurs at temperatures lower than those predicted by the theoretical calculations of Marinov [4], Park et al. [5], Li et al. [6], and Sivaramakrishnan et al. [7].

### 3.1.3. $C_2H_5OH + h = sC_2H_4OH + H_2$

Acetaldehyde time-histories measured during pyrolysis of 2%  $C_2H_5OH$ /Argon mixtures at 1 atm reported in [1] were used for determining the rate constant of the primary H-abstraction channel R3. The sensitivity of  $CH_3CHO$  time-history measured during one of the representative experiments is shown in Fig. 16(a). Also shown in Fig. 16(b) is the corresponding measurement plotted against the best-fit simulated time-histories and time-histories simulated by

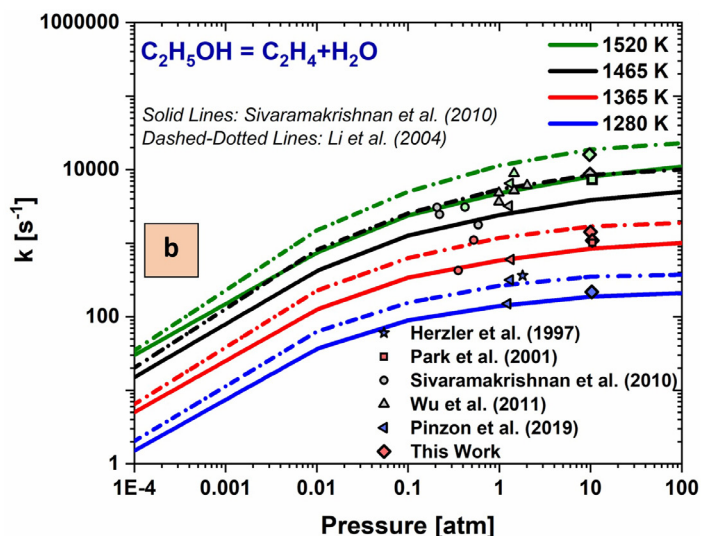
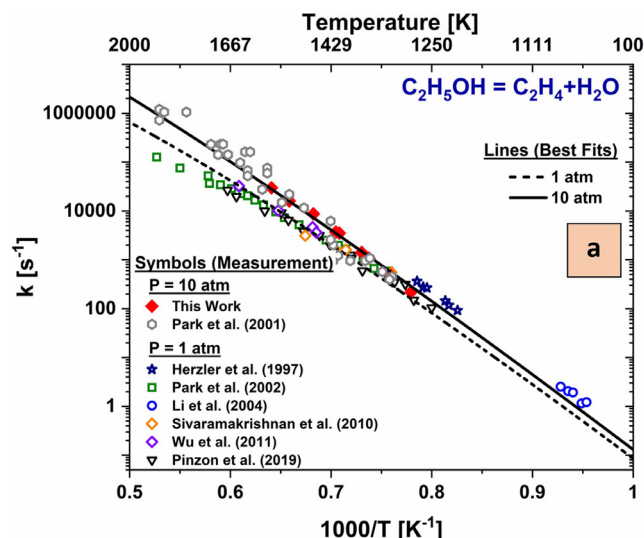


Fig. 14. (a) Rate constant expressions given by Eqs. (3) & 4 plotted against current and historical data. (b) Universal falloff curves for R2 calculated using the Sivaramakrishnan et al. [7] and Li et al. [6] expressions at four different temperatures. Symbols represent current and historical measurements of the rate constant.

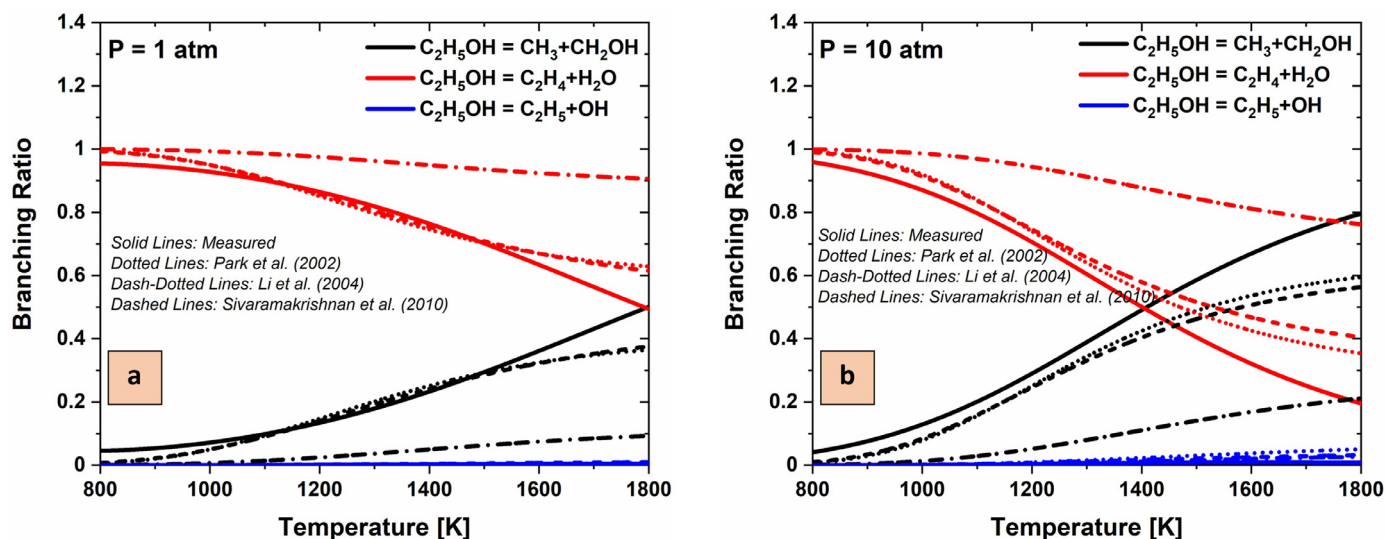


Fig. 15. Variation of branching ratios of the three channels of thermal decomposition of ethanol with temperature at (a) 1 atm, (b) 10 atm.

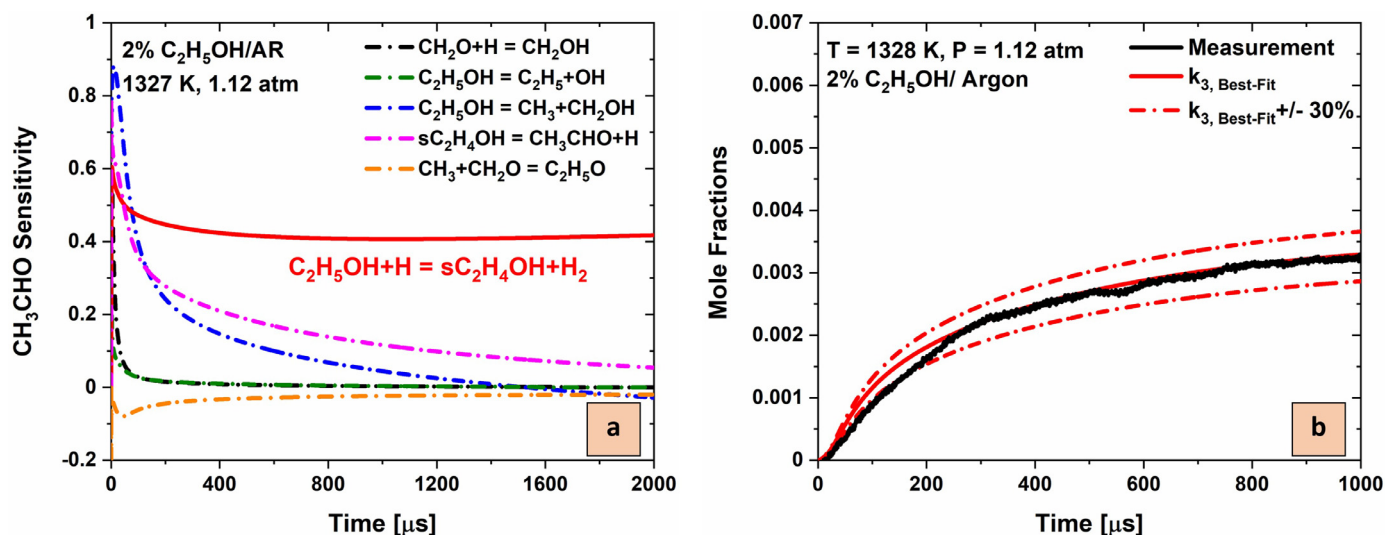


Fig. 16. (a) Sensitivity of  $\text{CH}_3\text{CHO}$  time-history during the pyrolysis of a 2%  $\text{C}_2\text{H}_5\text{OH}/\text{Argon}$  mixture at 1327 K, 1.12 atm. (b) Measured, and best-fit  $\text{CH}_3\text{CHO}$  time-histories obtained by adjusting  $k_3$ . The dash-dotted lines represent the  $\text{CH}_3\text{CHO}$  time-histories simulated by perturbing the best-fit rate constant by 30%. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

perturbing the best-fit rate constant by 30%. The sensitivity is dominated by R1 and R12 during the first 150  $\mu\text{s}$ , but this sensitivity diminishes at longer times.



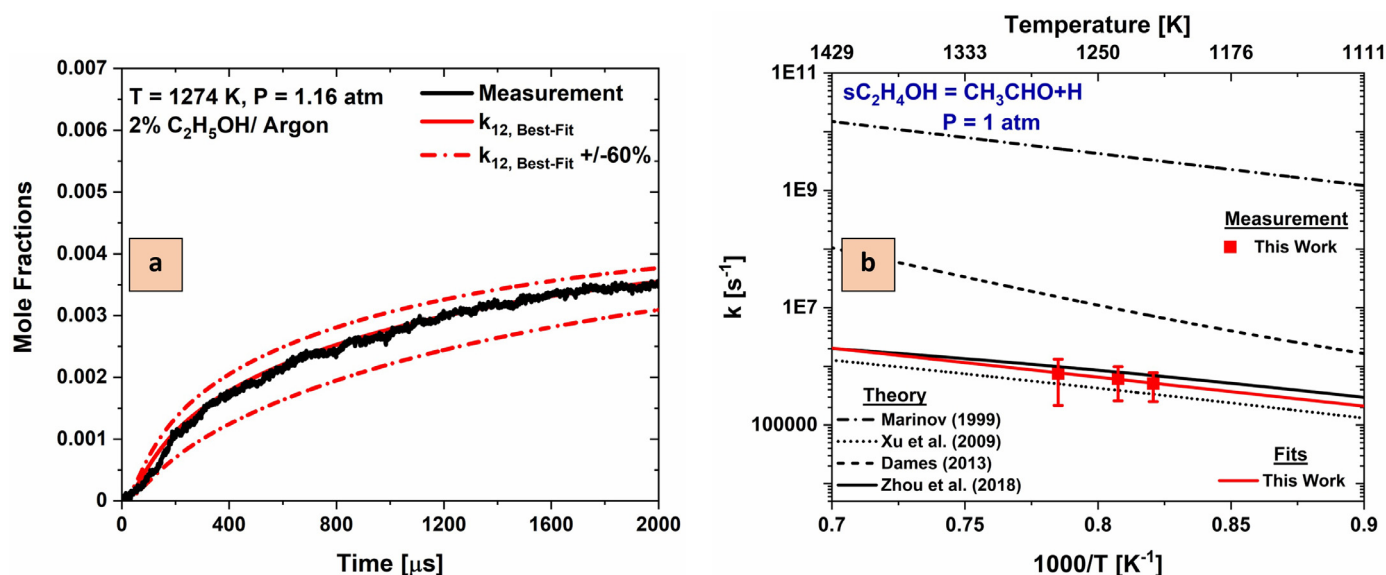
The rate constant of R1 at 1atm was measured in this work and has been already described in Section 3.1.1. However, no experimental studies of R12 have been reported in literature. Moreover, the recent theoretically calculated rate constants for R12, reported by Dames et al. [28] are an order of magnitude faster than the rate constant used in ARAMCO 3.0, while the rate constants used in the Marinov model are three orders of magnitude faster than the ARAMCO rate. The massive uncertainty in the rate constant of R12 will have a proportional impact on the uncertainty in  $k_3$  if it is determined using measured  $\text{CH}_3\text{CHO}$  time-histories. It was recognized that at initial reaction temperatures lower than 1280 K, the acetaldehyde time-histories are dominantly sensitive to R1 and R12 (fig. S.5.1 in Supplementary Material). The low temper-

ature  $\text{CH}_3\text{CHO}$  time-histories were consequently used to determine  $k_{12}$  using the procedure described in Sections 3.1.1 and 3.1.2. A representative measurement is shown in Fig. 17(a) along with the best fit simulated  $\text{CH}_3\text{CHO}$  time-history. Table A1 lists the rate constants measured at different temperatures along with corresponding uncertainties. The Arrhenius plot for R12 is shown in Fig. 17(b). The measured rate constants are 30% slower than the rate constant used in Aramco 3.0. The best fit to the measurements is given by Eq. (5). The overall 2- $\sigma$  uncertainty in  $k_{12}$  is  $\pm 55\%$ . Table S4 in the Supplementary Material lists the detailed breakdown of uncertainty in  $k_{12}$ .

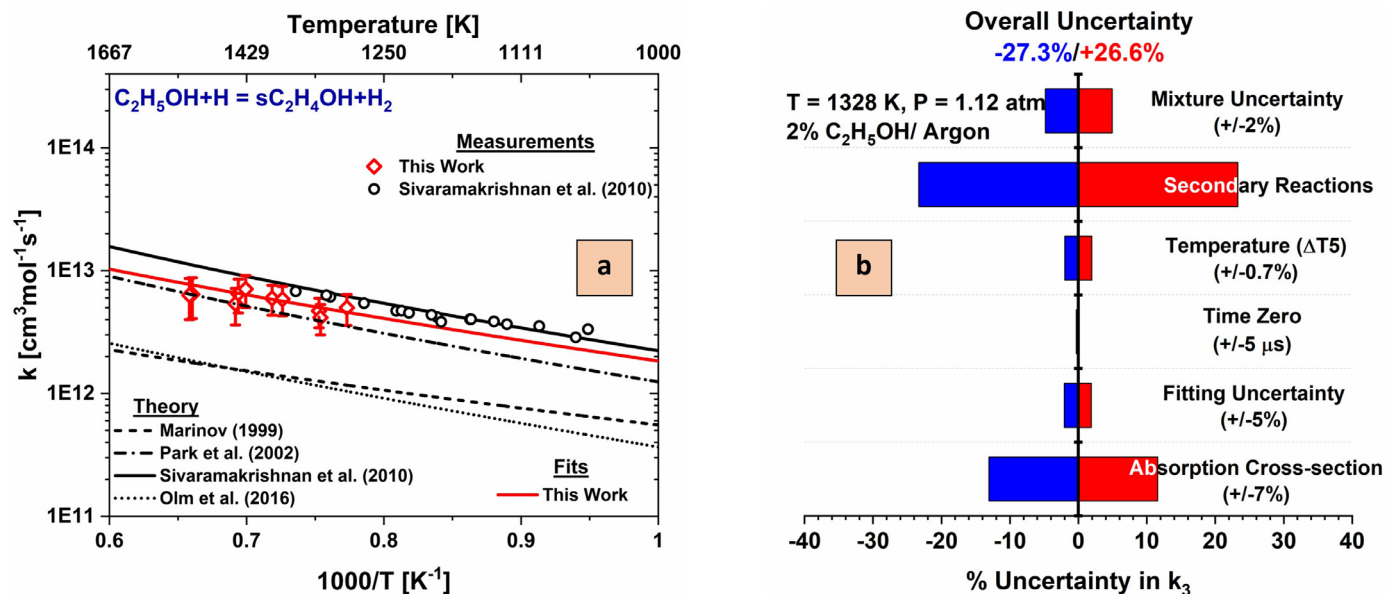
$$k_{12,1 \text{ atm}} = 5.817 \times 10^9 \exp\left(-\frac{11372}{T}\right) \text{ s}^{-1} \quad (5)$$

With  $k_{12}$  determined, the rate constants of the two dominant secondary reactions to which  $\text{CH}_3\text{CHO}$  time-histories are sensitive at longer times during pyrolysis at higher temperatures are known with uncertainty factors of 1.2 (R1) and 1.55 (R12). These time-histories were then used to determine  $k_3$  following the procedure





**Fig. 17.** (a) Measured, and best-fit  $\text{CH}_3\text{CHO}$  time-histories obtained by adjusting  $k_{12}$ ,  $1 \text{ atm}$ . The dash-dotted lines represent the  $\text{CH}_3\text{CHO}$  time-histories simulated by perturbing the best-fit rate constant by 30%. (b) Arrhenius plot for the rate constant of  $\text{sC}_2\text{H}_4\text{OH} = \text{CH}_3\text{CHO} + \text{H}$  at 1 atm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. 18.** (a) Arrhenius plot for the rate constant of  $\text{C}_2\text{H}_5\text{OH} + \text{H} = \text{sC}_2\text{H}_4\text{OH} + \text{H}_2$ . The best-fit curve is given by Eq. (6) (b) Uncertainty analysis for the measured  $k_3$  for the test case shown in Fig. 16.

that was adopted for  $k_1$ ,  $k_2$ , and  $k_{12}$ . A modified Aramco 3.0 model, with rate constants of R1, R2, and R12 replaced by the rate constant expressions given by Eqs. (1)–5 was used as the base model for fitting the  $\text{CH}_3\text{CHO}$  time-histories. Owing to the exclusive sensitivity of  $\text{CH}_3\text{CHO}$  time-histories to R3 between 500 and 2000  $\mu\text{s}$  (Fig. 16(a)), the time-histories measured in this interval were used with the fitting algorithm to determine the best-fit rate constants. The rate constants determined using the measured time-histories are compared against historical data and theoretical calculations on an Arrhenius plot in Fig. 18(a). The error bars represent the 2- $\sigma$  uncertainty in each measurement obtained by computing the root-mean-squared sum of the uncertainties stemming from various contributing sources.

The contribution of individual sources to the overall uncertainty is shown in Fig. 18(b). For the example case shown in Fig. 16, sec-

ondary reactions dominate the overall uncertainty ( $\sim 23\%$ ). Uncertainty in absorption cross-section is the second biggest contributor to the uncertainty ( $\sim 11\%$ ) followed by mixture uncertainty ( $\sim 5\%$ ). Owing to the small activation energy of the reaction, temperature contributes negligibly to the uncertainty. The overall uncertainty this case is  $-27.3\%/+26.6\%$ . The measured rate constants and the associated uncertainties for each experiment are tabulated in Table A1.

The values of  $k_3$  measured in this work are nearly 30% slower than the measurements of Sivaramakrishnan et al. [7]. As described in Section 1, Sivaramakrishnan et al. had measured the rate constant of the overall reaction  $\text{C}_2\text{H}_5\text{OH} + \text{D} = \text{Products}$ , instead of direct measurement of R3, which is the most dominant H-abstraction channel. For the sake of comparison with the measurements, the overall rate constant data of Sivaramakrishnan et al.



was multiplied by their theoretically calculated branching ratio for R3. The present measurements are in good agreement with the theoretically calculated rate constants of Park et al. [17], and almost an order of magnitude faster than the rate constants used in the Marinov and Olm et al. model. The best fit to the measurements is given by Eq. (6). Extrapolation of the best-fit to the data to lower temperatures (Fig. 19) also shows good agreement with the low temperature measurements of Aders & Wagner [16]. The sole measurement from Bansal & Freeman [15] at 425 K appears to be an outlier.

$$k_3 = 1.780 \times 10^7 T^{1.94} \exp\left(-\frac{1835}{T}\right) s^{-1} \quad (6)$$

### 3.2. Model performance

An updated kinetic model, henceforth referred to as the Stanford model, was constructed by replacing the rate constants of R1 (1 & 10 atm), R2 (1 & 10 atm), R3, and R12 by the rate constant expressions given by Eqs. (1)–6 in Aramco 3.0. The rate constants of thermal decomposition of  $CH_3CHO$  and  $CH_2O$ , H-abstraction from  $CH_3CHO$  and  $CH_2O$  by OH and H, were also replaced with the rate constants reported in recent studies conducted at Stanford by Wang and co-workers [21, 22, 29, 30], Vasudevan et al. [31], and Friedrichs et al. [32], and Yasunaga et al. [33]. Fig. S6.1 in the Supplementary Material suggests that updating the rate constants of reactions R1, R2, R3, and R12 strongly influences and improves the

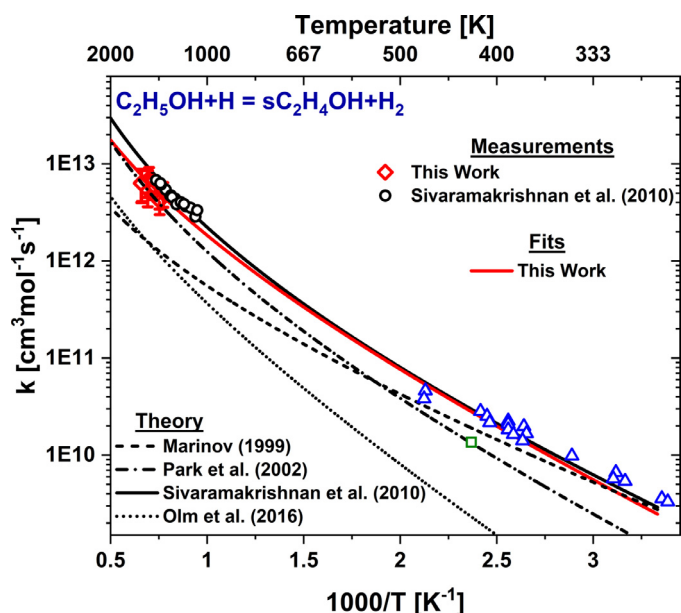


Fig. 19. Extended Arrhenius plot for the rate constant of  $C_2H_5OH + H = sC_2H_4OH + H_2$  comparing the extrapolation rate constant measured in this work to lower temperatures with historical data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

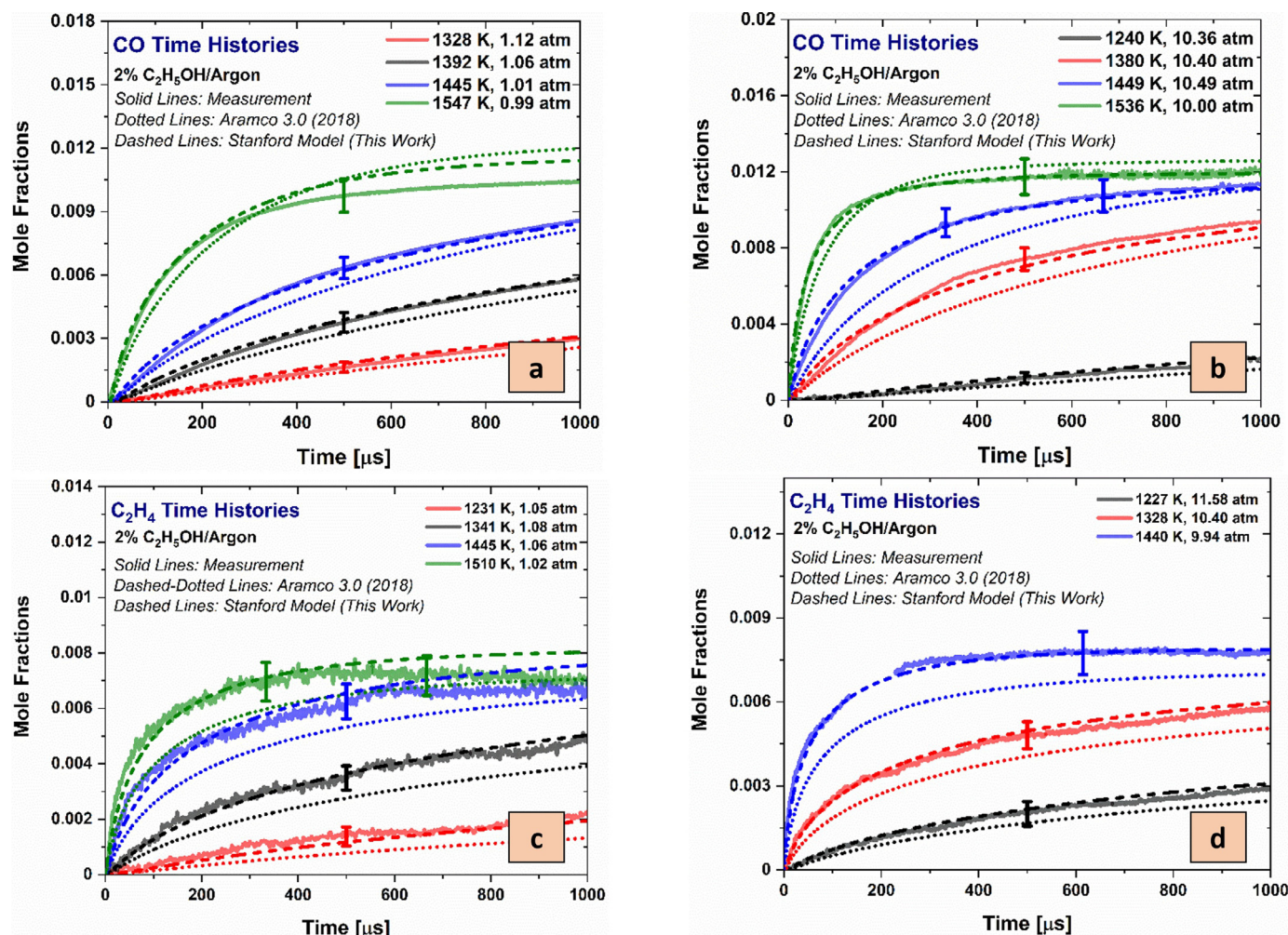
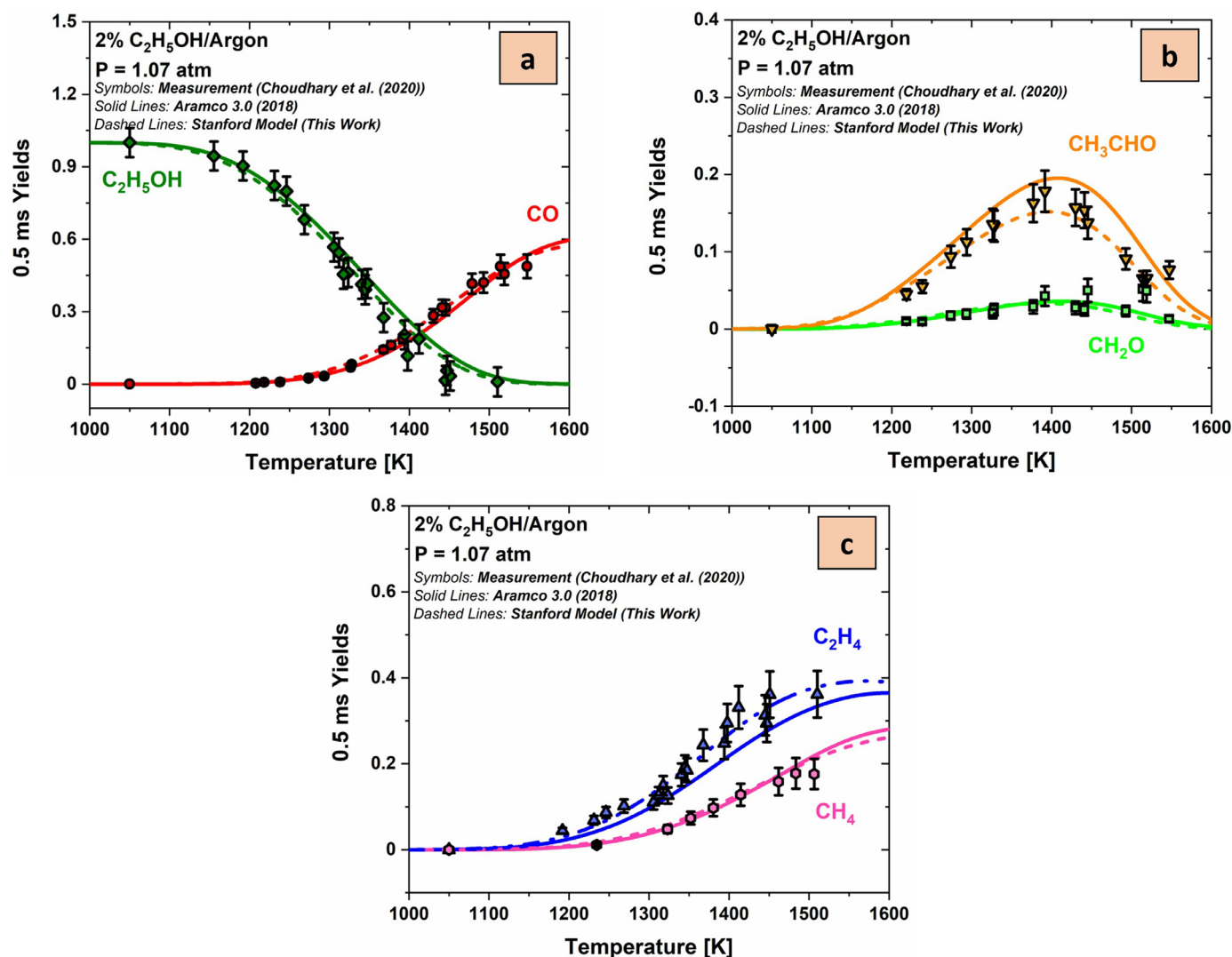


Fig. 20. Comparison of measured time-histories of CO ((a) 1 atm (b) 10 atm), and  $C_2H_4$  ((c) 1 atm (d) 10 atm) with the predictions of the Stanford model, and Aramco 3.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. 21.** Comparison of the measured early time (0.5 ms) pyrolysis yields of C<sub>2</sub>H<sub>5</sub>OH, CO (top left), CH<sub>2</sub>O, CH<sub>3</sub>CHO (top right), and C<sub>2</sub>H<sub>4</sub>, CH<sub>4</sub> (bottom) plotted against temperature for experiments conducted near 1.07 atm, with the prediction of the Stanford model and Aramco 3.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

predictive capability of the kinetic model. The influence of thermal decomposition and H-abstraction reactions of CH<sub>3</sub>CHO and CH<sub>2</sub>O is small but not insignificant and leads to further improvement in the predictive capability of the Stanford model. The performance of the Stanford model against shock tube ignition delay and species time-history data is described in the following subsection.

### 3.2.1. Comparison against shock tube speciation study of Choudhary et al. [1]

The pyrolysis experiments reported in the companion paper [1] were simulated using the Stanford model. The experiments were modeled as constant-pressure experiments (justification given in [1]). Fig. 20 shows the comparison of the measured time-histories of CO and C<sub>2</sub>H<sub>4</sub> forming during the pyrolysis of a 2% C<sub>2</sub>H<sub>5</sub>OH/Argon mixture at a selection of initial temperatures at 1 atm (left panel) and 10 atm (right panel), with the predictions of the Stanford model and Aramco 3.0. The time-histories predicted by the Stanford model show excellent agreement with the measured time-histories over the entire range of experiments studied in this work.

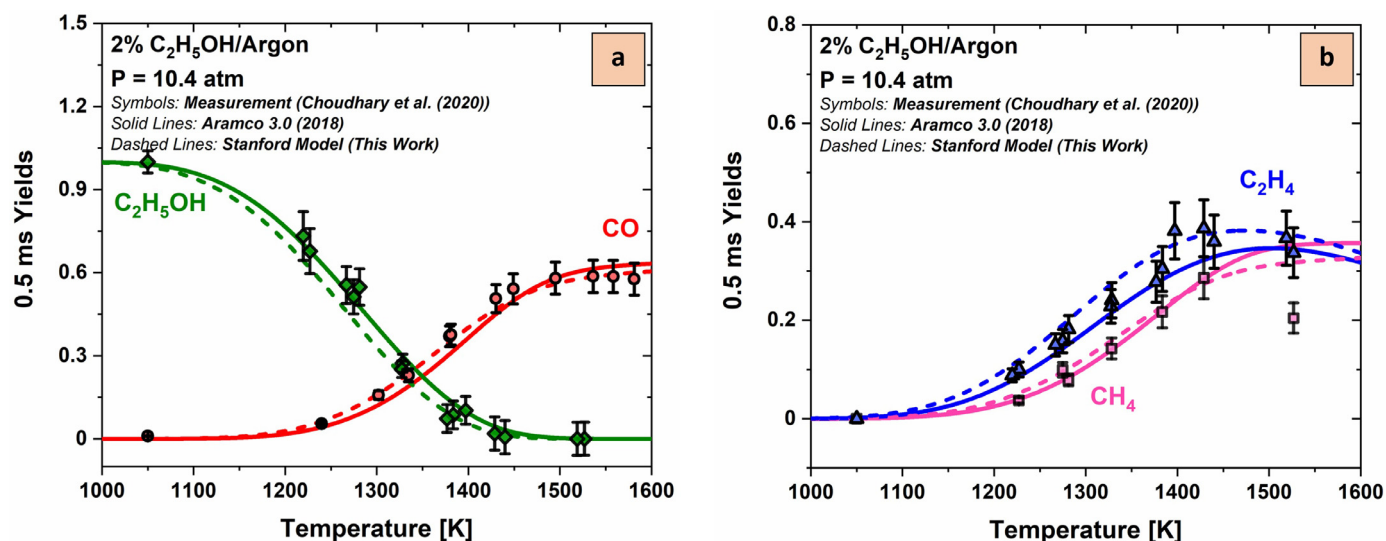
Figs. 21 and 22 show the performance of the Stanford model against 0.5 ms species yields, also reported in [1]. It is observed that the predictions of the Stanford model differ substantially from

those of Aramco 3.0, the base model on which it was built. A uniform improvement in the prediction of the yields of C<sub>2</sub>H<sub>5</sub>OH, and the key pyrolysis intermediates, and products, i.e., CO, C<sub>2</sub>H<sub>4</sub>, CH<sub>2</sub>O, CH<sub>4</sub>, and CH<sub>3</sub>CHO at 0.5 ms is observed for experiments performed at 1 atm. Similar improvement is observed in the predictions of the yields of C<sub>2</sub>H<sub>5</sub>OH, CO, C<sub>2</sub>H<sub>4</sub>, and CH<sub>4</sub> for experiments performed at 10 atm. These observations confirm that the Stanford model can successfully predict the temporal evolution of key species during the pyrolysis of ethanol. It is worth mentioning at this point that none of the five kinetic models evaluated in [1] could simultaneously reconcile all speciation measurements reported in [1].

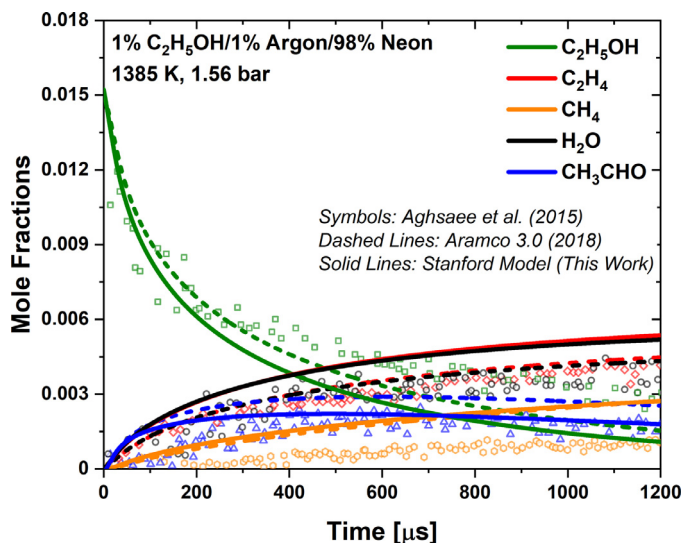
### 3.2.2. Comparison against shock tube speciation study of Aghsaei et al. [34], and Kiecherer et al. [11]

Aghsaei et al. [34] studied pyrolysis and oxidation of ethanol in shock tubes and laminar flames. The pyrolysis studies were conducted in a shock tube. Time-of-flight mass spectrometry (TOF-MS) was used to simultaneously measure the concentration of C<sub>2</sub>H<sub>5</sub>OH, H<sub>2</sub>O, CH<sub>4</sub>, C<sub>2</sub>H<sub>4</sub>, and CH<sub>3</sub>CHO in the reacting gas mixture, sampled every 10 μs from the shock tube test section. An example measurement from this study is shown in Fig. 23. This experiment was simulated using both Aramco 3.0, and the Stanford model, and the





**Fig. 22.** Comparison of the measured early time (0.5 ms) pyrolysis yields of C<sub>2</sub>H<sub>5</sub>OH, CO (left) and C<sub>2</sub>H<sub>4</sub>, CH<sub>4</sub> (right) plotted against temperature for experiments conducted near 10.4 atm, with the predictions of the Stanford model and Aramco 3.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. 23.** Comparison of the species time-histories predicted by the Stanford model and Aramco 3.0 with the measurements of Aghsaei et al. [34]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

species time histories predicted by both models are also shown in this figure.

Both models predict a faster decomposition of ethanol and correspondingly, a faster formation of pyrolysis products. The relatively good agreement between the measured CH<sub>3</sub>CHO time-histories and the prediction of the Stanford model, however, appears to be fortuitous. A possible reason for the observed discrepancy might be the absence of chaperon efficiency for Neon in both models for key pressure-dependent reactions, R1, R2, and R12. The experiments reported by Aghsaei et al. [34] utilized 1% C<sub>2</sub>H<sub>5</sub>OH/1% Argon/98% Neon as the test mixture. While plausible, it is unlikely that the identity of the bath gas would have such a significant impact on the rate of evolution of species.

The findings of Aghsaei et al. [34] are contradicted by the TOF-MS study of Kiecherer et al. [11]. While both studies utilized Neon as the bath gas, used similar test mixtures, spanned the same

range of test conditions, and were presented in the 2015 Combustion Symposium, their measured species time-histories show massive disagreements. The species time-histories measured by Kiecherer et al. at three different test conditions are shown in Fig. 24 (a–c). The predictions of the Stanford model show better agreement with the measurements for all species than Aramco. Among the measured species, C<sub>2</sub>H<sub>n</sub> (sum of C<sub>2</sub>H<sub>2</sub>, C<sub>2</sub>H<sub>4</sub>, and C<sub>2</sub>H<sub>6</sub>) time-histories show maximum deviation from the predictions of the Stanford model but still agree reasonably well within the scatter of the measurements and the uncertainty of the model. In general, it can be concluded that the Stanford model also shows a good agreement with the shock tube pyrolysis studies reported by other research groups.

### 3.2.3. Comparison against shock tube ignition delay time (IDT) measurements

Ignition delay times of two ethanol/oxygen/nitrogen mixtures were measured in the HPST facility at Stanford. Both mixtures were stoichiometric ethanol/oxygen mixtures, differing in the level of nitrogen dilution. The first mixture contained 6.54% C<sub>2</sub>H<sub>5</sub>OH/19.63% O<sub>2</sub>/N<sub>2</sub> while the second mixture contained 3.72% C<sub>2</sub>H<sub>5</sub>OH/11.17% O<sub>2</sub>/N<sub>2</sub>. The experiments were performed at 20 atm in the temperature range of 960–1200 K. A laser absorption diagnostic was used to measure the pre-shock concentration of ethanol in the test mixture at 3.41 μm. The diagnostic was also used for probing the evolution of C<sub>2</sub>H<sub>5</sub>OH and intermediates during ignition. An OH\* emission diagnostic was also used in conjunction with the absorption diagnostic for all ignition experiments. Pressure, OH\* emission, and absorbance at 3.41 μm were simultaneously recorded and used to infer the IDT.

Upon the onset of ignition, the pressure and emission signals show a rapid increase while there is a simultaneous rapid decay in the absorbance signal. The fastest rise in pressure and OH\* emission signals were extrapolated backwards in time to their respective baseline signal levels to infer the IDT. Similarly, the fastest decay in absorbance, extrapolated backwards in time to the pre-ignition absorbance level was also used for inferring the IDT. Table A2 lists the IDTs determined using all three signals, measured for both mixtures at different temperatures. Fig. 25 shows the time-histories measured during one of the experiments. The absorbance measured at 3.41 μm during oxidation has contributions from both ethanol and the intermediate species forming dur-



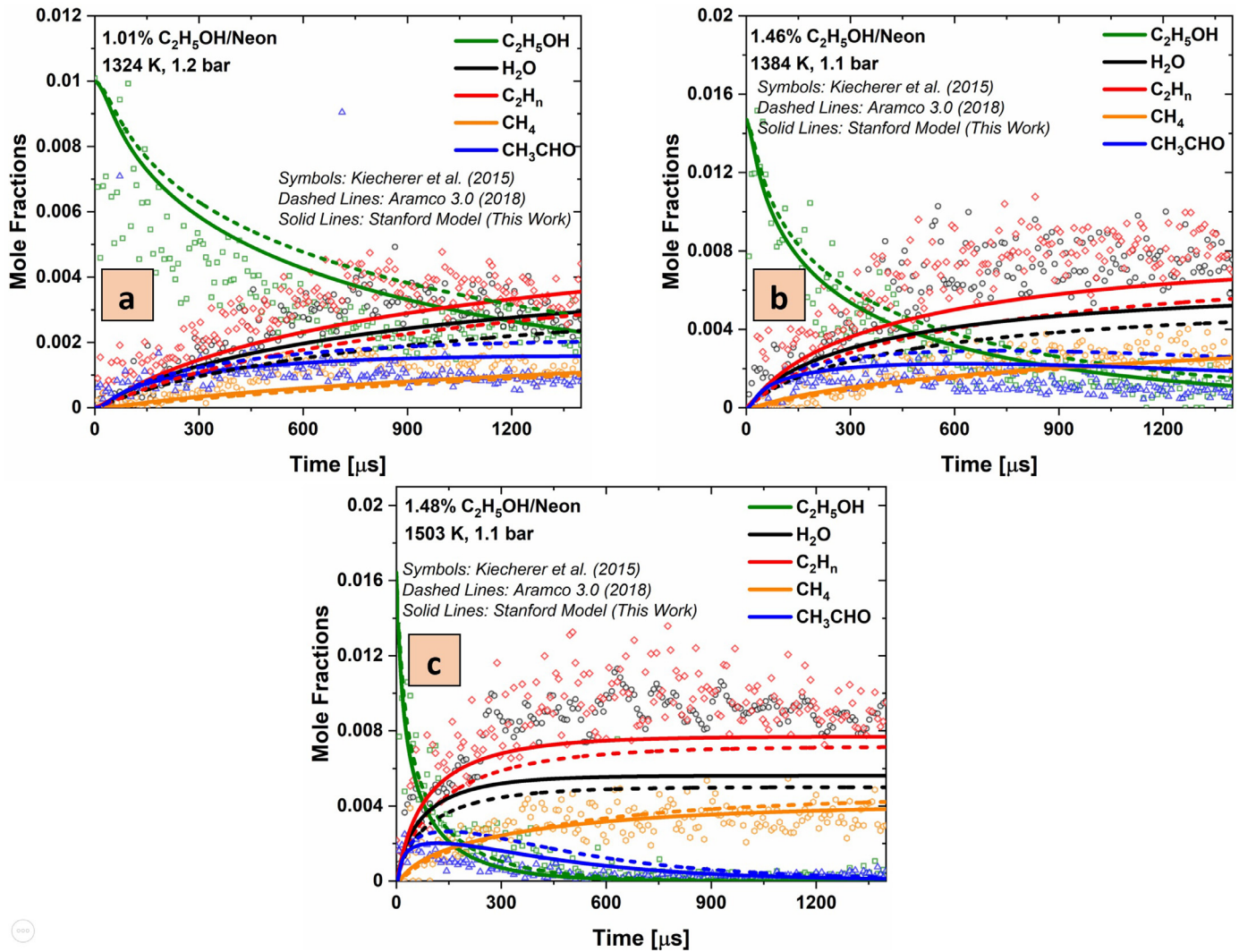


Fig. 24. Comparison of the species time-histories predicted by the Stanford model and Aramco 3.0 with the measurements of Kiecherer et al. [11]. Note that  $C_2H_n$  represents the composite sum of  $C_2H_2$ ,  $C_2H_4$ , and  $C_2H_6$ . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

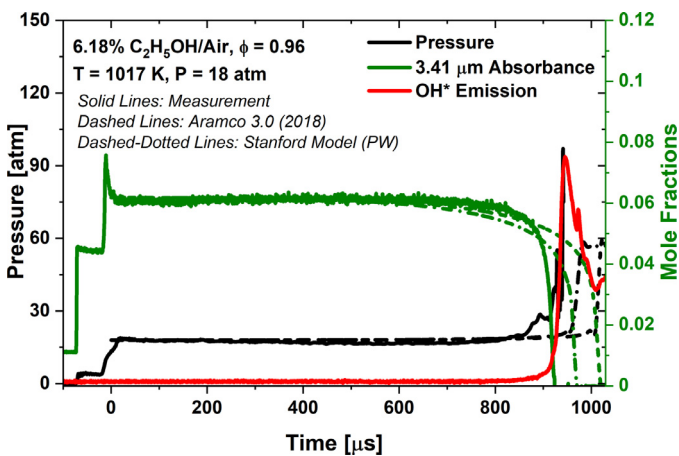


Fig. 25. Time-histories of pressure,  $OH^*$  emission and composite fuel + intermediate mole fractions measured during oxidation of a near stoichiometric ethanol/air mixture at 18 atm. Also shown are the predictions of the Stanford model and Aramco 3.0. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ing oxidation. It was assumed that all intermediate species, and ethanol have the same absorption cross-section. The known ab-

sorption cross-section of ethanol was then used with Beer's law (Eqn. (7)) to convert absorbance to mole fraction.

$$\alpha_v = -\ln\left(\frac{I_t}{I_0}\right)_v = \sum_i x_i n \sigma_{i,v}(T, P) L \quad (7)$$

Here,  $I_t$  and  $I_0$  are the transmitted and background intensities,  $\alpha_v$  is the absorbance,  $x_i$  is the mole fraction of the absorbing species at the probing wavelength,  $\sigma_{i,v}$  is the corresponding absorption cross-section at the probing wavelength,  $n$  is the number density of the test gas, and  $L$  is the path length of absorption that is same as the internal diameter of the shock tube. Thus, the time-history shown by the green curve in Fig. 25 is essentially a composite time-history of fuel and intermediate species. The comparison of this composite time history with the simulated ethanol time history is therefore, intended to be only qualitative.

Also shown in Fig. 25 are the ethanol, and pressure time-histories simulated using the Stanford model and Aramco 3.0. The experimental facility was modeled as a constant volume reactor in the simulations for the ignition experiments. At the test conditions of the example case, the IDT predicted by the Stanford model is in closer agreement with the measured IDT than the prediction of Aramco 3.0. The measured IDTs for both mixtures are plotted against temperature, along with the measurements of Mathieu et al. [35], Laich et al. [36], and Lee et al. [37], in Fig. 26. The mea-

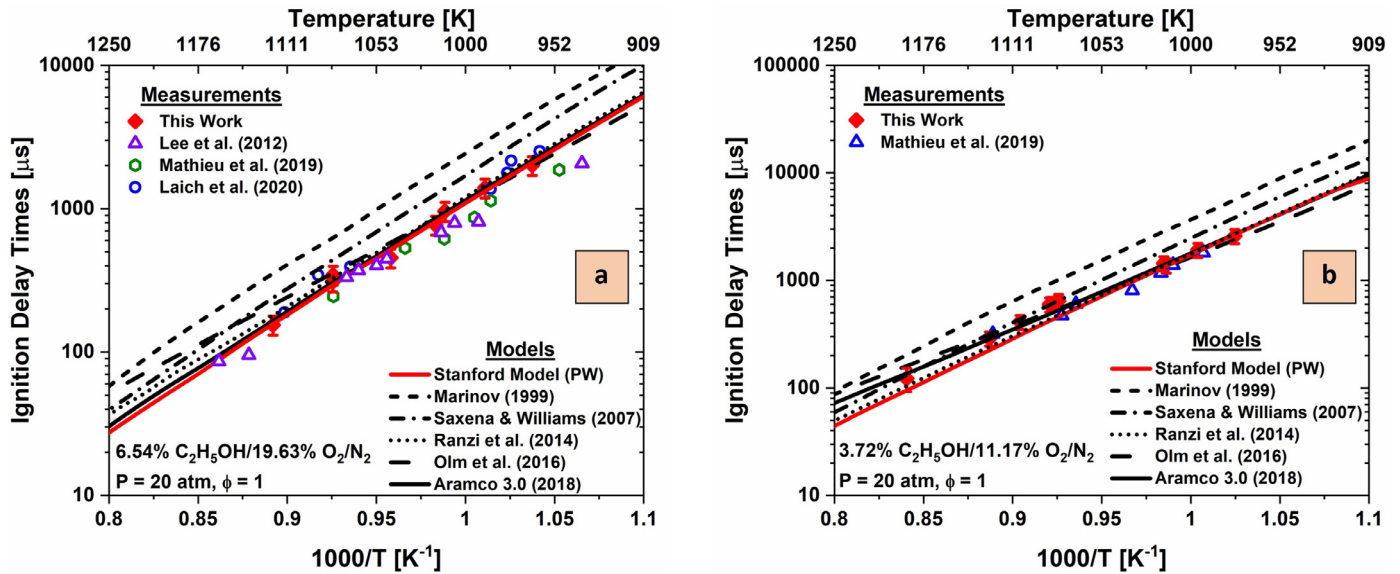


Fig. 26. Comparison between the current measurements of Ignition Delay Times with other recent measurements, and the predictions of kinetics models for (a) Stoichiometric, undiluted ethanol/air mixture, (b) Stoichiometric, ethanol/oxygen mixture further diluted in nitrogen at 20 atm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

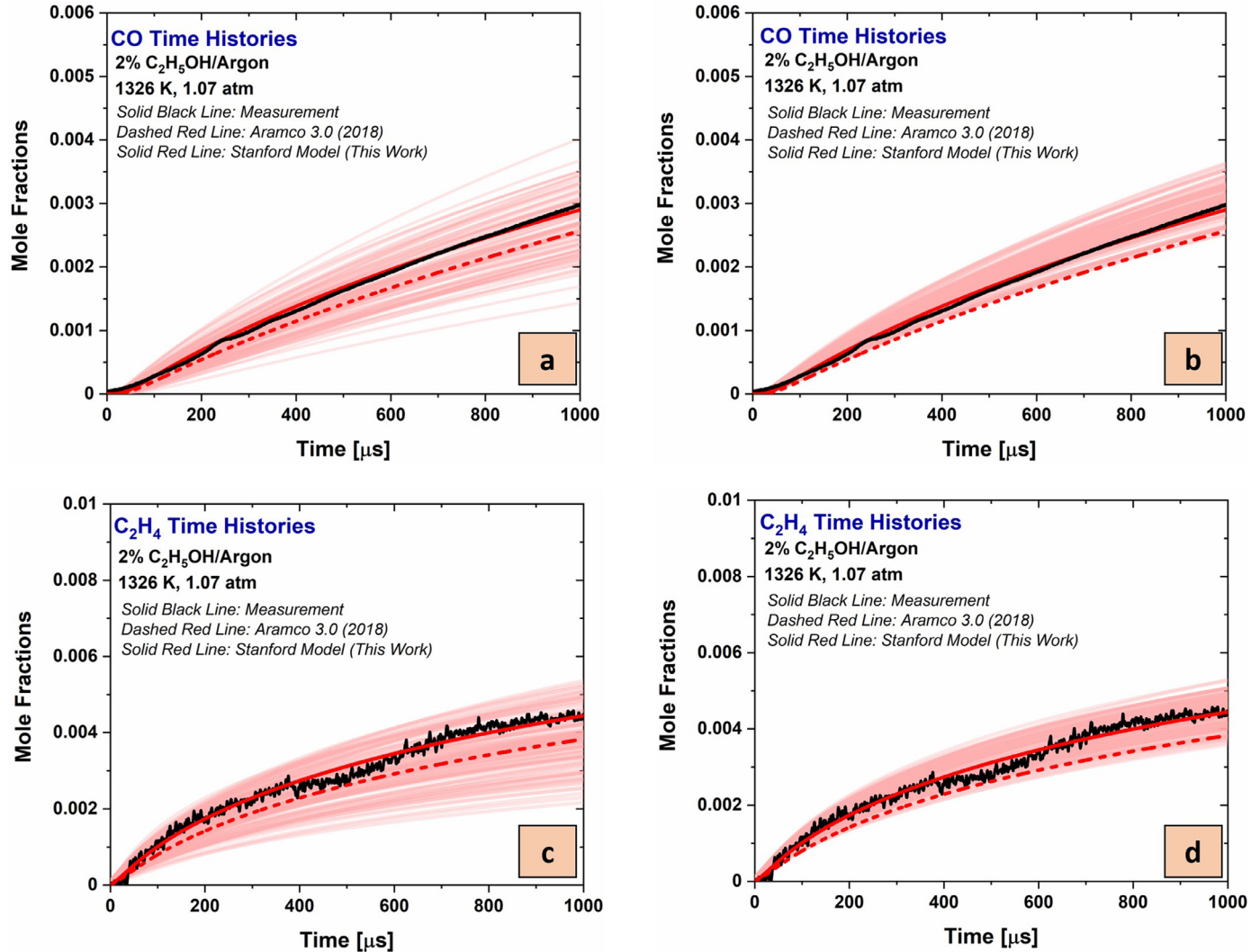
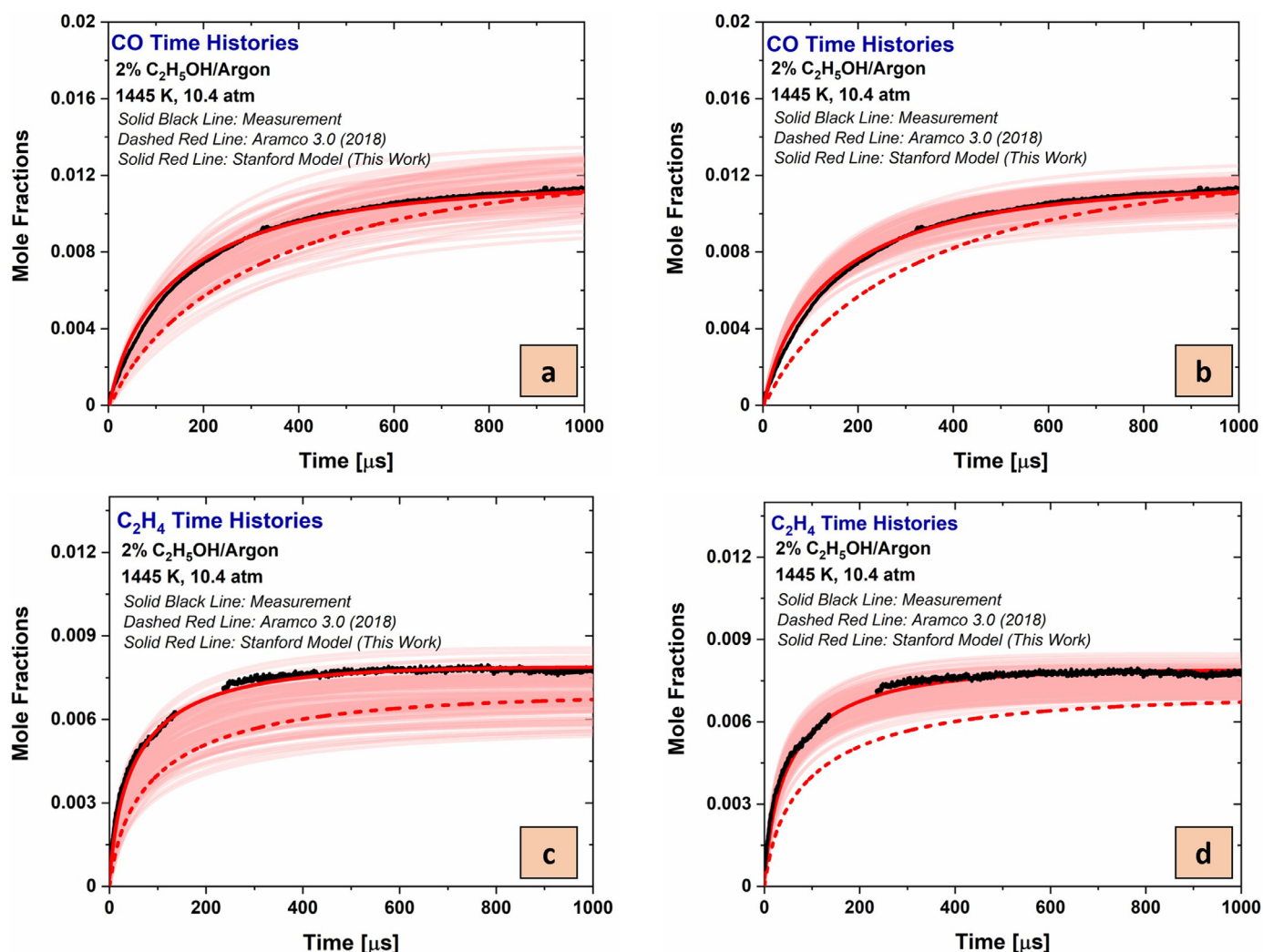


Fig. 27. (a-b) Monte Carlo sampling of the predicted CO time-histories during pyrolysis of 2% C<sub>2</sub>H<sub>5</sub>OH in Argon at 1326 K, 1.07 atm generated using (a) Aramco 3.0 and (b) Stanford model as the base model. (c-d) Monte Carlo sampling of the predicted C<sub>2</sub>H<sub>4</sub> time-histories at the same test conditions generated using (c) Aramco 3.0 and (d) Stanford model as the base model. Also shown are the measured time-histories (reproduced from [1]), and nominal predictions of both models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. 28.** (a-b) Monte Carlo sampling of the predicted CO time-histories during pyrolysis of 2%  $\text{C}_2\text{H}_5\text{OH}$  in Argon at 1445 K, 10.4 atm generated using (a) Aramco 3.0 and (b) Stanford model as the base model. (c-d) Monte Carlo sampling of the predicted  $\text{C}_2\text{H}_4$  time-histories at the same test conditions generated using (c) Aramco 3.0 and (d) Stanford model as the base model. Also shown are the measured time-histories (reproduced from [11]), and nominal predictions of both models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sured IDTs were scaled to 20 atm by assuming an inverse dependence on pressure ( $\text{IDT} \propto P^{-1}$ ). Also plotted are the IDTs predicted by the Stanford model and five recent kinetic models. The present measurements show excellent agreement with the measurements of Mathieu et al., Laich et al., and Lee et al. for mixture 1, and with the measurements of Mathieu et al. for mixture 2. The IDTs predicted by the Stanford model, Aramco 3.0, and the Ranzi et al. model [18], show excellent agreement with the current measurements. Out of these three models, the Stanford model shows the best agreement with the measurements for both mixtures; the differences between the models, however, are small. The minor differences between the predictions of the Stanford model and Aramco 3.0 suggest that the IDTs are weakly sensitive to the reactions that have a dominating influence during pyrolysis, i.e. R1-R4, and R12. Therefore, modification of the rate constants of these reactions using Eqs. (1)-6 only has a moderate impact on the predicted IDTs.

### 3.3. Model uncertainty

The discussion in the preceding subsections demonstrates that the predictions of the Stanford model show good agreement not only with multiple speciation studies conducted during pyrolysis of ethanol in shock tubes, but also with multiple shock tube IDT stud-

ies. It is also critical to assess the uncertainty in these predictions. The procedure adopted is a modification of the Monte Carlo sampling approach that was followed to quantify the uncertainty in the measured rate constants stemming from secondary reactions.

For quantification of the uncertainty in predictions of time-histories forming during pyrolysis, all reactions of  $\text{C}_2\text{H}_5\text{OH}$ ,  $\text{CH}_2\text{O}$ ,  $\text{CH}_3\text{CHO}$ , and  $\text{C}_2\text{H}_4$  that do not involve oxygen, or the radicals forming during oxidation ( $\text{HO}_2$ , peroxy radicals etc.) were considered. The uncertainty factors for the rate constants of each of these reactions were sourced from their corresponding studies referenced in Aramco 3.0. An uncertainty factor of 2 was assumed for all reactions for which the uncertainty factors were not mentioned in their respective sources. A set of one hundred sample perturbed models were generated using both Aramco 3.0 and the Stanford model as the base model, by multiplying the rate constants of each reaction in the list by a factor sampled randomly from their corresponding uncertainty range, with the latter assumed to be uniformly distributed. For the perturbed models generated using the Stanford model as the base model, the rate constants, and the uncertainty factors of R1, R2, R3, and R12, reported in this work were used. The full list of reactions along with their uncertainty factors, and references are tabulated in section S7 of the Supplementary Material.



Each experiment reported in [1] was simulated using the one hundred perturbed models generated using both Aramco 3.0 and the Stanford models as the base models. The results of these simulations are shown for a representative 1 atm experiment (Fig. 27), and 10 atm experiment (Fig. 28). In both figures, the time-histories of CO and C<sub>2</sub>H<sub>4</sub> simulated using the perturbed models with Aramco as the base model are shown on the left panels, while the time-histories simulated using the Stanford model as the base model are shown on the right panels. Also shown are the nominal predictions of the base models and the measured time-histories. The impact of reduction in the uncertainty of the key ethanol reactions is immediately obvious. For the representative low-pressure case shown in Fig. 27, the predicted CO time-histories lie within 60% of the nominal prediction for the sampling performed using Aramco as the base model. In comparison, the time-histories predicted by the Stanford model lie within 25% of the nominal prediction of the Stanford model. The C<sub>2</sub>H<sub>4</sub> time-histories also show a similar reduction in the prediction uncertainty. Fig. 28 shows the results of the Monte Carlo sampling for a representative 10 atm experiment. Akin to the 1 atm case, the sampled time-histories predicted by the Stanford model span a narrower range about the nominal prediction than Aramco (10% versus 25% for CO time-histories and 20% versus 30% for C<sub>2</sub>H<sub>4</sub> time-histories). The relatively small change in the prediction uncertainty can be attributed to the lower absolute sensitivity that the species time-histories exhibit during pyrolysis at 10 atm than at 1 atm.

## Conclusions

The deficiencies in the kinetic models describing the pyrolysis of ethanol were identified in [1] by means of species time-history measurements in shock tubes, enabled by laser absorption spectroscopy. The discrepancies were attributed to the inaccurate rate constants used for a small set of reactions including R1: C<sub>2</sub>H<sub>5</sub>OH(+M) = CH<sub>2</sub>OH+CH<sub>3</sub>(+M), R2: C<sub>2</sub>H<sub>5</sub>OH(+M) = C<sub>2</sub>H<sub>4</sub>+H<sub>2</sub>O(+M), and R3: C<sub>2</sub>H<sub>5</sub>OH+H = sC<sub>2</sub>H<sub>4</sub>OH+H<sub>2</sub>. Additional experiments were conducted to determine the rate constant of R1 near 1 atm. CO time-histories measured during the pyrolysis of 0.5% C<sub>2</sub>H<sub>5</sub>OH/Argon mixtures enabled the determination of k<sub>1</sub> with an overall 2-σ uncertainty of +/- 21%. The CO time-histories measured during high-pressure pyrolysis experiments in [1] were used for determination of k<sub>1</sub> at 10 atm with an overall 2-σ uncertainty of +/- 28%. Addition of CH<sub>4</sub> to C<sub>2</sub>H<sub>5</sub>OH/Argon mixtures boosted the sensitivity of C<sub>2</sub>H<sub>4</sub> time-histories to R2 and resulted in the suppression of the sensitivity of C<sub>2</sub>H<sub>4</sub> time-histories to secondary reactions. The C<sub>2</sub>H<sub>4</sub> time-histories measured during the pyrolysis of 1% C<sub>2</sub>H<sub>5</sub>OH/1% CH<sub>4</sub>/Argon mixtures were used for determining the rate constant of R2 at 10 atm within tight uncertainty bounds of +/- 31%. Additionally, the low-temperature CH<sub>3</sub>CHO time-histories reported in [1] were then used to infer the rate constant of R12: sC<sub>2</sub>H<sub>4</sub>OH = CH<sub>3</sub>CHO + H. With the rate constants of R1, R2, and R12 updated, the acetaldehyde time-histories measured at higher initial temperatures were used for determining the rate constant of R3 with an overall 2-σ uncertainty of +/- 28%. The rate constants of R3 and R12 reported in this work are the first direct measurements of these rate constants. While enabling improved theoretical modeling of the rate constants of these reactions, these measurements will be instrumental in identifying critical remaining uncertainties in the knowledge base of these reactions and will serve as a guide for future experimental studies.

An updated chemical kinetic model (the Stanford model) was generated by replacing the rate constants of R1, R2, R3, and R12 in Aramco 3.0 with the rate constants measured in this work. Rate constants of unimolecular decomposition and H-abstraction reactions of CH<sub>2</sub>O and CH<sub>3</sub>CHO were also replaced with the rate constants reported in more recent experimental studies. The pyrolysis experiments reported in [1] were simulated using the Stanford

model, and its predicted species time-histories were found to be in excellent agreement with the measurements over the entire range of experimental conditions. The model was able to simultaneously reconcile with the measured time-histories of all species; a significant improvement over the five chemical kinetic models considered in [1]. The Stanford model also exhibited improved agreement with the TOF-MS studies of Aghsaee et al. [34], and Kiecherer et al. [1], within the measurement uncertainties of both the studies. Ignition delay times of two stoichiometric C<sub>2</sub>H<sub>5</sub>OH/O<sub>2</sub>/N<sub>2</sub> mixtures, differing in the extent of nitrogen dilution were also measured near 20 atm to provide additional targets for the validation and evaluation of the Stanford model. The measured IDTs were found to be in excellent agreement with the recent studies of Laich et al. [36], Lee et al. [37], and Mathieu et al. [35]. The predictions of the Stanford model and Aramco 3.0 were also found to be in excellent agreement with the measurements. While the differences

**Table A1**

Summary of rate constant measurements.

| Temperature (K)   | Pressure (atm) | K (s <sup>-1</sup> ) | 2-σ Uncertainty (%) |      |
|-------------------|----------------|----------------------|---------------------|------|
|                   |                |                      | -                   | +    |
| <b>R1 (1 atm)</b> |                |                      |                     |      |
| 1469              | 1.09           | 2094                 | -23.3               | 22.9 |
| 1316              | 1.16           | 143                  | -19.7               | 19.7 |

(continued on next page)

**Table A1 (continued)**

| Temperature (K)    | Pressure (atm) | K (s <sup>-1</sup> ) | 2-σ Uncertainty (%) |      |
|--------------------|----------------|----------------------|---------------------|------|
| 1234               | 1.18           | 21                   | -19.7               | 19.7 |
| 1334               | 1.14           | 145                  | -19.8               | 20.1 |
| 1367               | 1.13           | 339                  | -21.1               | 21.1 |
| 1504               | 1.05           | 3166                 | -26.0               | 25.2 |
| 1293               | 1.20           | 73                   | -19.7               | 19.3 |
| 1440               | 1.11           | 1348                 | -21.7               | 21.4 |
| <b>R1 (10 atm)</b> |                |                      |                     |      |
| 1335               | 10.49          | 570                  | -28.2               | 28.8 |
| 1240               | 10.36          | 59                   | -27.9               | 28.0 |
| 1379               | 10.11          | 1728                 | -27.8               | 28.4 |
| 1449               | 10.00          | 7439                 | -33.4               | 30.5 |
| 1536               | 10.00          | 27,600               | -28.6               | 30.0 |
| 1581               | 10.00          | 39,173               | -28.3               | 29.4 |
| 1430               | 10.49          | 5214                 | -28.2               | 28.9 |
| 1381               | 10.29          | 1841                 | -28.0               | 28.6 |
| 1495               | 10.38          | 18,616               | -28.4               | 29.4 |
| 1558               | 10.00          | 42,032               | -30.0               | 28.9 |
| 1302               | 10.40          | 289                  | -28.3               | 28.2 |
| <b>R2 (10 atm)</b> |                |                      |                     |      |
| 1369               | 9.98           | 1417                 | -30.6               | 31.2 |
| 1412               | 9.94           | 3511                 | -30.8               | 31.3 |
| 1419               | 9.95           | 3703                 | -31.0               | 31.1 |
| 1465               | 9.84           | 8717                 | -42.8               | 37.5 |
| 1560               | 9.58           | 29,830               | -42.4               | 47.6 |
| 1518               | 9.73           | 16,013               | -36.8               | 38.5 |
| 1356               | 10.45          | 1087                 | -30.7               | 31.6 |
| 1283               | 10.28          | 218                  | -34.8               | 36.0 |
| 1317               | 10.79          | 481                  | -39.3               | 34.6 |
| <b>R3</b>          |                |                      |                     |      |
| 1328               | 1.12           | 4.71E+12             | -27.3               | 26.6 |
| 1392               | 1.06           | 5.96E+12             | -27.1               | 27.7 |
| 1445               | 1.01           | 5.38E+12             | -32.8               | 33.6 |
| 1514               | 1.01           | 6.41E+12             | -36.4               | 36.8 |
| 1441               | 1.02           | 6.51E+12             | -30.5               | 31.1 |
| 1519               | 1.00           | 6.32E+12             | -36.8               | 37.0 |
| 1430               | 1.03           | 7.09E+12             | -29.2               | 29.3 |
| 1377               | 1.09           | 5.86E+12             | -26.9               | 27.0 |
| 1327               | 1.11           | 4.15E+12             | -27.7               | 27.8 |
| 1294               | 1.13           | 5.00E+12             | -28.4               | 28.5 |
| <b>R12</b>         |                |                      |                     |      |
| 1218               | 1.18           | 510,280              | -50.7               | 51.8 |
| 1274               | 1.16           | 755,644              | -71.5               | 74.5 |
| 1238               | 1.17           | 618,273              | -58.2               | 58.8 |

**Table A2**

Summary of IDT measurements.

| Mixture 1 (6.54% C <sub>2</sub> H <sub>5</sub> OH/19.63% O <sub>2</sub> /N <sub>2</sub> ) |                 |                          |              |            |       |
|-------------------------------------------------------------------------------------------|-----------------|--------------------------|--------------|------------|-------|
| Pressure (atm)                                                                            | Temperature (K) | Ignition Delay Time (μs) |              |            | Φ     |
|                                                                                           |                 | Pressure                 | OH* Emission | Absorption |       |
| 22.35                                                                                     | 996             | 931.2                    | 932          | 921.6      | 0.95  |
| 21.07                                                                                     | 1080.2          | 324                      | 327.2        | 328.4      | 0.9   |
| 18.59                                                                                     | 1011.6          | 1028                     | 1032         | 1043       | 0.92  |
| 19.49                                                                                     | 989.1           | 1414                     | 1434         | 1451       | 0.91  |
| 17                                                                                        | 1017.13         | 905.6                    | 906          | 918.4      | 0.92  |
| 16.15                                                                                     | 1043.75         | 561.6                    | 562.4        | 567.2      | 0.92  |
| 18.94                                                                                     | 1121            | 156                      | 163.2        | 163.6      | 0.88  |
| 21.08                                                                                     | 1080.4          | 290.4                    | 293.6        | 294.4      | 0.9   |
| 23.16                                                                                     | 963.9           | 1725                     | 1733         | 1700       | 0.92  |
| Mixture 2 (3.72% C <sub>2</sub> H <sub>5</sub> OH/11.17% O <sub>2</sub> /N <sub>2</sub> ) |                 |                          |              |            |       |
| Pressure (atm)                                                                            | Temperature (K) | Ignition Delay Time (μs) |              |            | Φ     |
|                                                                                           |                 | Pressure                 | OH* Emission | Absorption |       |
| 21.7                                                                                      | 976             | 2366                     | 2377         | 2327       | 0.94  |
| 21.02                                                                                     | 996             | 1836                     | 1867         | 1820       | 0.89  |
| 20.18                                                                                     | 1085            | 614.8                    | 618          | 614.4      | 0.93  |
| 20.21                                                                                     | 1086            | 596.4                    | 603.2        | 600        | 0.93  |
| 20.85                                                                                     | 1106            | 431.2                    | 414.4        | 435.2      | 0.87  |
| 19.4                                                                                      | 1015.4          | 1516                     | 1525         | 1524       | 0.918 |
| 21.43                                                                                     | 1080.2          | 610.8                    | 622.8        | 615.6      | 0.91  |
| 18.89                                                                                     | 1013.9          | 1449                     | 1456         | 1464       | 0.947 |
| 19.9                                                                                      | 1127.4          | 303.2                    | 304          | 312        | 0.928 |
| 19.47                                                                                     | 1189.85         | 136                      | 136.8        | 141.6      | 0.89  |

were small, the Stanford model exhibited better agreement with the measurements in general, than Aramco 3.0. Reduction in the uncertainty of rate constants of critical reactions measured in this work also had a proportional impact on the uncertainty in the model predictions. A Monte Carlo sampling study was conducted using both Aramco 3.0 and the Stanford model to assess the uncertainties in the predictions of both models. A two-fold reduction in the uncertainty was observed in the CO, and C<sub>2</sub>H<sub>4</sub> time-histories simulated using the Stanford model compared to Aramco 3.0.

In conclusion, the study of ethanol pyrolysis described in this paper and in [1] highlights the important role that multi-species time-history studies enabled by laser absorption spectroscopy and shock tubes play in uncovering the deficiencies in our knowledge of the chemical kinetics of even simple molecules. Starting from the generation of high-quality validation targets in the form of species time-histories, followed by the identification of key reactions and precise and accurate measurement of their rate constants, this study has resulted in the development of an accurate, and well-validated kinetic model for ethanol pyrolysis. The low uncertainty associated with the predictions of this model make it an ideal benchmark for evaluating the quality of future modeling and experimental studies of ethanol pyrolysis.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2021.111554](https://doi.org/10.1016/j.combustflame.2021.111554).

## Appendix

### Tables A1 and A2

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